“ASK Modem Performance under Variable Noise”

A PROJECT REPORT

Submitted by

Anshika Chauhan	**(DL.AI.U4AID24106)**
Kalyani Agarawal	**(DL.AI.U4AID24114)**
Madesh KK	**(DL.AI.U4AID24120)**
Parima Verma	**(DL.AI.U4AID24125)**

in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY (B. Tech) IN ARTIFICIAL INTELLIGENCE AND DATA SCIENCE (AIDS)

Under the Guidance of

Dr. Abhishek Kumar (Assistant Prof.(SDCS))

Dr. Mrittunjoy Guha Majumdar (Assistant Prof.(MSA))

Submitted to

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

AMRITA SCHOOL OF ARTIFICIAL INTELLIGENCE

FARIDABAD – 121002

December 2025

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ARTIFICIAL INTELLIGENCE FARIDABAD

BONAFIDE CERTIFICATE

This is to certify that this project report entitled "ASK Modem Performance under Variable Noise" is the **bonafide work** of the following students, who carried out the project work under my supervision:

- **Anshika Chauhan** (DL.AI.U4AID24106)
- **Kalyani Agarawal** (DL.AI.U4AID24114)
- **Madesh KK** (DL.AI.U4AID24120)
- **Parima Verma** (DL.AI.U4AID24125)

SIGNATURE

Dr. Abhishek Kumar	**Dr.Mrittyunjoy Guha Majumdar**
Assistant Professor	Assistant Professor
School of AI	School of AI

Amrita Vishwa Vidyapeetham, Faridabad

AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ARTIFICIAL INTELLIGENCE FARIDABAD

DECLARATION BY THE CANDIDATES

We hereby declare that the project report entitled "**ASK Modem Performance under Variable Noise**" submitted by us in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** is the record of the project work carried out by us under the guidance of "**Dr. Abhishek Kumar(SDCS) Dr. Mrittunjoy Guha Majumdar(MSA)**". This work has not formed the basis for the award of any degree, diploma, associateship, fellowship, or title in this or any other University or institution of higher learning.

SIGNATURES

Anshika Chauhan
DL.AI.U4AID24106

Kalyani Agarawal
DL.AI.U4AID24114

Madesh KK
DL.AI.U4AID24120

Parima Verma
DL.AI.U4AID24125

ACKNOWLEDGEMENT

This project work would not have been possible without the contribution and support of many people. We take immense pleasure in expressing our profound gratitude to our honorable Chancellor, **Sri Mata Amritanandamayi Devi**, for her blessings and for being a constant source of inspiration.

We express our sincere gratitude to our Principal, **Dr. Lakshmi Mohandas**, Amrita School of Computing and Engineering, for providing the necessary facilities and continuous support to gain valuable education and learning experience.

We would like to express our heartfelt gratitude to our supervisors, **Dr. Abhishek Kumar** and **Dr. Mrittunjoy Guha Majumdar**, for their personal involvement, invaluable guidance, and constant encouragement throughout the course of this project work.

We are thankful to the Project Supervisor, Review Panel Members, and the entire faculty of the Department of Computer Science & Engineering for their constructive criticism and valuable suggestions, which have been a rich source for improving the quality of this work.

Anshika Chauhan
DL.AI.U4AID24106

Kalyani Agarawal
DL.AI.U4AID24114

Madesh KK
DL.AI.U4AID24120

Parima Verma
DL.AI.U4AID24125

Abstract

This project presents a comprehensive design, simulation, and performance analysis of a digital communication system employing Amplitude Shift Keying (ASK), specifically the On-Off Keying (OOK) modulation scheme. Implemented within the Software Defined Radio (SDR) framework of GNU Radio Companion and validated via MATLAB, the study constructs a robust model that decomposes a transceiver into functional sub-systems, including stochastic source generation, carrier modulation, and coherent demodulation. By simulating an Additive White Gaussian Noise (AWGN) channel, the study introduces stochastic behavior to a deterministic system, allowing for a rigorous analysis of signal integrity.

System performance is quantitatively validated using Monte Carlo simulations to compare simulated Bit Error Rate (BER) and spectral bandwidth against theoretical mathematical models. The results demonstrate the efficacy of replacing hardware-dependent circuits with reconfigurable software blocks, enabling real-time visualization through virtual instrumentation. Ultimately, this research highlights the versatility of SDR in rapid prototyping and its accuracy in predicting the behavior of physical communication systems under varying interference levels.

Contents

List of Figures

List of Tables

List of Symbols

Primary Symbols

A_c	Carrier amplitude
f_c	Carrier frequency
R_b	Bit rate
E_b	Energy per bit
N_0	Noise power spectral density
B_T	Transmission bandwidth
sps	Samples per symbol

Signal Modeling Symbols

$s(t)$	Transmitted modulated signal
$m(t)$	Binary message signal (baseband digital stream)
$c(t)$	Carrier signal
$r(t)$	Received signal after passing through the channel
$n(t)$	Additive noise component

Performance and Probability Symbols

γ	Optimal decision threshold for decoding bits
P_e, P_b	Probability of error or Bit Error Rate (BER)
σ	Noise amplitude voltage (standard deviation) in AWGN model
$Q(\cdot)$	Q-function (for theoretical BER calculation)
η	Power efficiency
P_{avg}	Average power of the signal

System and Sampling Symbols

F_s, f_s	System sampling rate (e.g., 32 kHz)
T_b	Bit period (duration of one bit)
N_{cycles}	Number of carrier cycles per symbol
τ_g	Group delay introduced by filters
Δt	Cumulative processing delay to be compensated

1 Introduction

The rapid evolution of modern telecommunications has necessitated a paradigm shift from rigid, hardware-centric architectures to flexible, software-defined ecosystems. This project, *"ASK Modem Performance under Variable Noise,"* explores this transition by implementing a complete digital communication chain within the **Software Defined Radio (SDR)** framework. By leveraging **GNU Radio Companion** for dynamic system modeling and **MATLAB** for theoretical validation, this study bridges the gap between abstract mathematical communication theory and practical signal processing implementation.

The core objective of this research is to design, simulate, and rigorously analyze an **Amplitude Shift Keying (ASK)** transceiver system, specifically utilizing **On-Off Keying (OOK)** modulation. In real-world environments, communication signals are invariably corrupted by stochastic disturbances. To replicate these conditions, this project models the channel as a Linear Time-Invariant (LTI) system introducing **Additive White Gaussian Noise (AWGN)**, enabling a robust investigation into signal integrity and system resilience.

Key technical contributions of this work include:

- **Dynamic System Modeling:** Decomposing the transceiver into functional subsystems —stochastic source generation, carrier modulation, and coherent demodulation—to visualize the complete signal processing pipeline.
- **Stochastic Performance Analysis:** Quantifying system reliability through **Bit Error Rate (BER)** analysis, using **Monte Carlo simulations** to evaluate performance across a spectrum of Signal-to-Noise Ratios (SNR).
- **Theoretical Validation:** Correlating simulated spectral efficiency and error probability with fundamental information theory benchmarks (e.g., the Q-function), demonstrating the fidelity of SDR-based prototyping.

This report documents the design methodology, simulation parameters, and critical findings of the project, highlighting how MSA (Modeling, Simulation, and Analysis) principles can be effectively applied to predict the behavior of physical communication systems without the prohibitive costs of hardware prototyping.

1.1 Literature Survey

1.1.1 Historical Evolution of Software Defined Radio

The paradigm of wireless communication has undergone a radical transformation with the advent of Software Defined Radio (SDR). The conceptual foundation was laid in the early 1990s by Joseph Mitola III, who coined the term "Software Radio" to describe a transceiver where signal processing is defined in software rather than fixed hardware [1]. This architecture promised a "universal radio" capable of inter-operating with differing standards solely through software reconfiguration [2]. Initial implementations were largely restricted to military applications, such as the DARPA "SpeakEasy" project, due to the prohibitive cost of Analog-to-Digital Converters (ADCs) and Digital Signal Processors (DSPs) [3]. However, the evolution of semiconductor technology has since democratized SDR, making it a viable platform for commercial and academic research [4]. As noted by Tuttlebee, this shift represents a fundamental move from hardware-centric to software-centric engineering, reducing development cycles and enabling rapid prototyping [5].

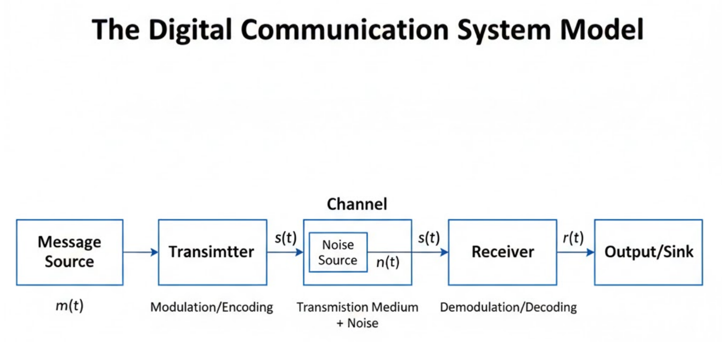


Figure 1: Time-domain representation of signal components. The message signal $m(t)$ is used to modulate the carrier $c(t)$ at frequency f_c, resulting in the transmitted waveform $s(t)$.

1.1.2 Digital Modulation and ASK Theory

At the physical layer, the efficiency of a communication system is dictated by its modulation scheme. Amplitude Shift Keying (ASK), particularly its binary form On-Off Keying (OOK), remains one of the earliest and simplest forms of digital modulation [6]. While less power-efficient than Phase Shift Keying (PSK) or Frequency Shift Keying (FSK), ASK offers significant advantages in terms of receiver simplicity [7]. According to Shannon's Information Theory, the capacity of a channel is fundamentally limited by bandwidth and signal-to-noise ratio, a principle that governs the design of all modulation schemes [8]. Recent studies continue to explore ASK for low-power IoT applications and optical communications where implementation complexity is a critical constraint [9]. The trade-offs between spectral efficiency and power efficiency in ASK are well-documented in standard texts by Proakis and Salehi [10].

1.1.3 The Role of GNU Radio in Modern Engineering

The emergence of open-source tools has accelerated the adoption of SDR in education and industry. GNU Radio, created by Eric Blossom, provides a flowgraph-oriented framework that abstracts complex signal processing mathematics into modular blocks [11]. This "visual programming" approach allows engineers to model complete transceiver chains without writing low-level C++ or VHDL code [12]. Research by Ulversoy highlights that GNU Radio has become the de facto standard for software radio research, enabling the simulation of everything from simple AM radios to complex LTE stacks [13]. Furthermore, its integration with hardware peripherals like the USRP (Universal Software Radio Peripheral) has bridged the gap between simulation and real-world transmission [14]. Educational case studies have demonstrated that students grasp abstract communication concepts significantly faster when using GNU Radio compared to traditional textbook methods [15].

1.1.4 Stochastic Modeling and BER Analysis

A critical aspect of communication system design is analyzing performance under stochastic interference. The Additive White Gaussian Noise (AWGN) channel model is universally accepted as the standard baseline for evaluating digital modems [16]. As detailed by Sklar, the Bit Error Rate (BER) serves as the primary metric for reliability, typically following a ”waterfall” curve that decays exponentially with increasing Signal-to-Noise Ratio (SNR) [17]. For non-coherent ASK, the theoretical probability of error is derived from the Rician distribution of the envelope detector outputs [18]. Recent literature emphasizes the importance of Monte Carlo simulations in verifying these theoretical error bounds, particularly when using software-defined architectures [19]. The fidelity of such simulations is contingent on accurate noise generation and proper synchronization, challenges that are frequently discussed in SDR implementation papers [20].

1.1.5 Current Trends and Future Directions

The scope of SDR and digital modulation extends far beyond basic communication. Modern applications include Cognitive Radio, where systems intelligently detect and utilize idle spectrum [21]. Integrating Machine Learning with SDR is a burgeoning field, allowing radios to classify modulation types and adapt to jamming automatically [22]. In the context of 5G and 6G, SDR is pivotal for testing massive MIMO and beamforming algorithms before silicon fabrication [23]. Despite these advancements, the fundamental principles of Nyquist sampling and matched filtering remain the bedrock of all digital systems [24]. As the industry moves toward ”Virtual Radio Access Networks” (vRAN), the skills developed in modeling basic SDR systems become directly transferable to next-generation telecommunications infrastructure [25].

1.2 Objectives

The primary goal of this project is to design, model, and analyze a digital communication system using GNU Radio. The specific objectives, aligned with the Course Outcomes, are:

- **To Design and Implement Digital Modulation (SDR CO3):** Create a functional flowgraph for an Amplitude Shift Keying (ASK) transceiver using Software Defined Radio platforms.
- **To Apply Modeling Techniques (MSA CO1, CO3):** Utilize computer-based modeling to represent the transmitter, channel, and receiver as a unified dynamic system.
- **To Analyze System Performance (MSA CO2, SDR CO1):** Evaluate signal attributes such as Bit Error Rate (BER) and spectral efficiency by simulating the system under varying noise conditions (AWGN).
- **To Validate Simulation with Theory (MSA CO4):** Compare the simulated results (waveforms and constellations) with theoretical mathematical models to verify the accuracy of the software design.

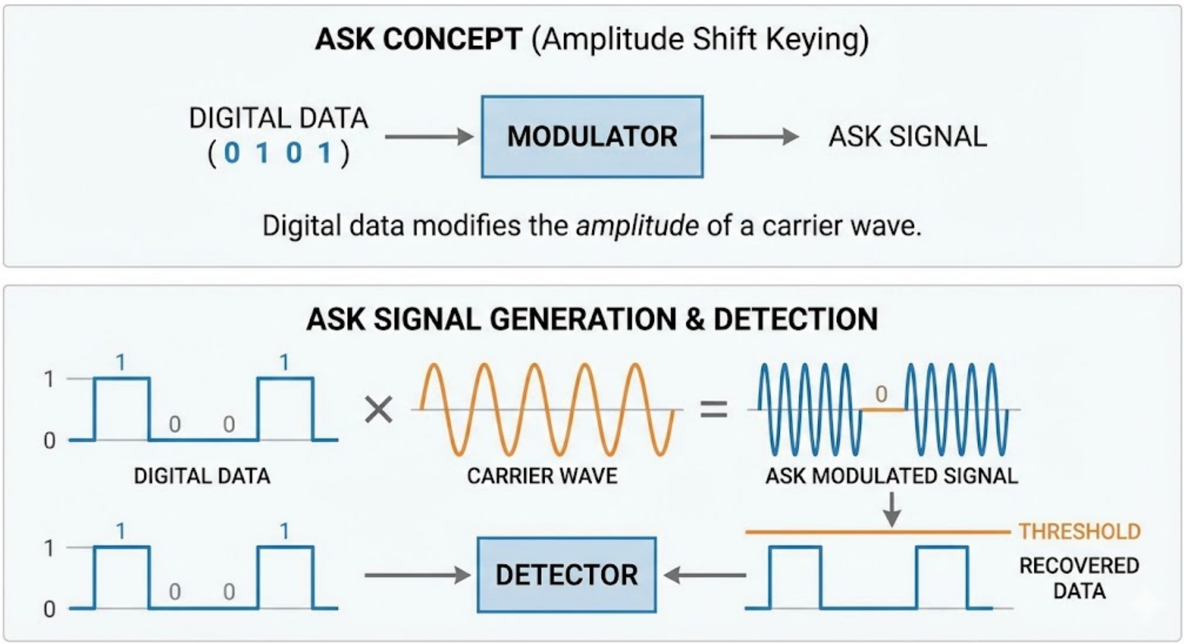


Figure 2: ASK Signal Generation and Non-Coherent Detection Process

1.3 Theoretical Background

A. Principle of ASK (Amplitude Shift Keying)

Amplitude Shift Keying (ASK) is a form of digital modulation where the amplitude of the carrier wave is varied in accordance with the baseband digital stream. In its simplest form, known as On-Off Keying (OOK), the carrier is transmitted to represent a binary '1' and suppressed to represent a binary '0'. This technique is widely used in low-frequency RF applications due to its simplicity.

- **Transmitted Signal Representation:**

$$s(t) = \begin{cases} A_c \cos(2\pi f_c t), & \text{for binary '1'} \\ 0, & \text{for binary '0'} \end{cases}$$

- **Optimal Decision Rule:** In an Additive White Gaussian Noise (AWGN) channel, the optimal decision threshold γ is defined as:

$$\gamma = \frac{A_c}{2}$$

- **Application:** Given a carrier amplitude $A_c = 1$:

$$\gamma = \frac{1}{2} = 0.5$$

  This threshold minimizes the Bit Error Rate (BER) by bisecting the distance between the two signaling states.

2 System Modeling Methodology

To ensure a robust analysis, the project followed a structured simulation workflow based on system dynamics and modeling principles. This methodology decomposes the communication chain into functional stages, ensuring that the software simulation accurately reflects theoretical behaviors.

1. **System Representation**
   The transceiver chain was modeled as a dynamic system, where the state of the system evolves over time. The model is defined by:
   - **Inputs:** A stochastic binary data stream (0s and 1s) generated by a random source[1].
   - **Process:** The modulation of a carrier wave and transmission through a medium.
   - **Disturbances:** The introduction of Additive White Gaussian Noise (AWGN) to simulate real-world channel interference[2].
2. **Mathematical Formulation**
   Before simulation, governing equations were derived to link the time-domain signal to theoretical performance.
   - The **Modulated Signal** $s(t)$ was defined as the product of the message and carrier:
   $$s(t) = m(t) \times A_c \cos(2\pi f_c t) \tag{1}$$
   - The **Received Signal** $r(t)$ was modeled to include channel effects:
   $$r(t) = s(t) + n(t) \tag{2}$$
   where $n(t)$ represents the noise component[4].
3. **Flowgraph Construction**
   A GNU Radio block diagram was constructed to represent the functional signal flow. This step involved translating the mathematical model into discrete processing blocks, such as the Signal Source for carrier generation and Multiply blocks for mixing[5]. Key parameters, including gain and interpolation factors, were identified to ensure signal integrity throughout the chain.
4. **Parameter Initialization**
   Simulation parameters were explicitly defined and justified to satisfy the Nyquist Criterion ($F_s \geq 2f_{max}$) to prevent aliasing[6]:
   - **Sample Rate:** Set to 32 kHz to provide sufficient resolution for the 2 kHz carrier[7].
   - **Carrier Frequency:** Set to 2 kHz to allow for clear visualization in the time domain[8].
   - **Samples per Symbol:** Calculated as 32 to ensure smooth waveform representation[9].
5. **Simulation Execution**
   Time-domain and frequency-domain simulations were executed using virtual instrumentation. Time Sinks were utilized to capture the waveform envelopes before and after modulation, while Frequency Sinks (FFT) were used to observe spectral occupancy and sideband generation[10].

6. **Monte Carlo Analysis**
   A dynamic analysis was performed to evaluate the system's robustness against stochastic interference. The Noise Power (N_0) was varied across a defined range, simulating $10,000$ bits per SNR point to observe steady-state Bit Error Rates (BER)[7]. This high iteration count ensures the statistical significance of the results by minimizing the impact of transient noise spikes.

7. **Validation and Performance Verification**
   To verify the integrity of the model, simulated results were compared against theoretical benchmarks:

   - **Bit Error Rate (BER):** The simulated BER curve extracted from MATLAB was plotted against the theoretical Q-function curve ($P_e = Q(\sqrt{\frac{E_b}{N_0}})$) to confirm model accuracy[8].
   - **Bandwidth:** The theoretical bandwidth ($BW \approx 2R_b$) was cross-referenced with the spectral width observed in the simulation[11].
   - **Power Efficiency:** The energy distribution between the carrier and sidebands was analyzed to confirm standard ASK behavior.

8. **Dynamic Analysis**
   The system's dynamic response was analyzed by varying the Noise Power (N_0). By adjusting the noise amplitude in the AWGN block, the system was tested under different Signal-to-Noise Ratios (SNR), allowing for the observation of transient errors and steady-state Bit Error Rates (BER)[12].

9. **Validation & Optimization**
   The final model was validated by verifying that the simulated BER conformed to the theoretical probability curve:

$$P_e \approx \frac{1}{2}e^{-\frac{2E_b}{N_0}} \tag{3}$$

   Discrepancies were minimized by optimizing the receiver's Threshold value to distinctively separate logic '1' from logic '0'[14].

B. Mathematical Model (MSA CO2)

The communication system is modeled as a dynamic system where the input message $m(t)$ is transformed into a transmitted signal $s(t)$ and received as $r(t)$. The transmitter and channel are modeled as Linear Time-Invariant (LTI) subsystems, as the operations (multiplication by a constant carrier, addition of noise) satisfy the principles of superposition. However, the non-coherent detection (thresholding) at the receiver introduces a nonlinear element essential for digital decision-making.

Modulated Signal: The output signal $s(t)$ is the product of the binary message signal $m(t)$ and the carrier signal $c(t)$:

$$s(t) = m(t) \times A_c \cos(2\pi f_c t) \tag{4}$$

Where:

- $m(t) = 1$ for binary '1' and 0 for binary '0'.

- f_c is the carrier frequency (set to 2 kHz in this simulation).

Channel Model: The channel introduces Additive White Gaussian Noise (AWGN). The received signal $r(t)$ is:

$$r(t) = s(t) + n(t) \tag{5}$$

Where $n(t)$ represents the Gaussian noise component with configurable amplitude.

Demodulation (Threshold Detection): The receiver recovers the original bits by comparing the received energy against a threshold. If the voltage level exceeds the threshold (e.g., 100 mV), it is decoded as a '1'; otherwise, it is a '0'.

C. Simulation Parameters (MSA CO4)

The following system parameters define the dynamic behavior of the model in GNU Radio:

- **Sample Rate (F_s):** 32 kHz. This defines the time resolution of the simulation.
- **Bit Rate (R_b):** 1 kHz. This determines the speed of data transmission.
- **Samples per Symbol (SPS):** 32. Calculated as F_s/R_b, ensuring sufficient data points per bit for accurate waveform representation.
- **Symbol Table:** $0, 1$. This confirms the implementation of On-Off Keying (OOK).
- **Noise Source:** Gaussian distribution with variable amplitude to test system robustness.

D) Expected Theoretical Relationships

To validate the "Simulation, Modeling, and Analysis" (MSA) objectives, the following theoretical relationships quantify the system's performance. These baselines will be compared against the simulated values obtained from the GNU Radio flowgraph.

1. Bandwidth Requirement (BW)

For an Amplitude Shift Keying (ASK) signal, the transmission bandwidth is directly proportional to the bit rate. For the main lobe of the spectrum:

$$BW \approx 2R_b$$

where R_b is the Bit Rate.

In this project: With a Bit Rate of 1 kHz, the expected theoretical bandwidth is 2 kHz.

2. Power Efficiency (η)

ASK is generally less power-efficient than other modulation schemes because a significant portion of the transmitted power is contained in the carrier frequency (which carries no information) rather than the sidebands.

Average Power (P_{avg}): Since the carrier is only transmitted for binary '1' (assuming equal probability of 0s and 1s):

$$P_{\text{avg}} = \frac{1}{2} \cdot 2A_c^2 = A_c^2$$

where A_c is the carrier amplitude.

3. Bit Error Rate (BER) Probability

The performance of the system in the presence of Additive White Gaussian Noise (AWGN) is defined by the probability of error P_e. For non-coherent ASK detection (using thresholding as done in this flowgraph):

$$P_e \approx \frac{1}{2} e^{-\frac{2E_b}{N_0}}$$

Relationship: As the Noise Amplitude N_0 increases, the Signal-to-Noise Ratio (SNR) decreases, leading to an exponential increase in the Bit Error Rate.

Simulation Check: We expect the BER to be near zero when the noise voltage is low (e.g., 0.1) and rise significantly as noise approaches the signal amplitude.

4. Data Rate and Sampling Relationship

To ensure the simulation successfully models the continuous-time signal (Nyquist Criterion):

$$F_s \geq 2f_{\text{max}}$$

In this project: The Sample Rate is 32 kHz, which is well above the carrier frequency of 2 kHz, ensuring no aliasing occurs during the simulation.

3 Simulation Design in GNU Radio

3.1 Flowgraph Description

The system is implemented as a unidirectional data pipeline in GNU Radio Companion (GRC). The signal processing chain is divided into five distinct stages, modeling the lifecycle of the signal from bit generation to error analysis.

1. Digital Source & Baseband Processing (Transmitter)

The first stage involves generating the raw digital information and preparing it for transmission.

- **Bit Generation (Random Source):** The process begins with the Random Source block, which generates a stochastic stream of binary data (0s and 1s). This ensures the system is tested against unpredictable data patterns.
- **Symbol Mapping (Chunks to Symbols):** The raw bits are passed to the Chunks to Symbols block. This performs the initial mapping for On-Off Keying (OOK), converting the binary integer stream into floating-point symbols, where a '0' bit maps to 0.0V and a '1' bit maps to 1.0V.
- **Upsampling (Repeat):** To satisfy the Nyquist criterion and ensure smooth waveform representation, the discrete symbols are upsampled. The Repeat block replicates each symbol 32 times ($SPS = 32$). This expands the data from a Bit Rate (R_b) of 1 kHz to the system Sample Rate (F_s) of 32 kHz.

2. Analog Modulation (Mixing Stage)

This stage shifts the baseband signal to the passband frequency for transmission.

- **Carrier Generation (Signal Source):** An independent Signal Source block generates a continuous cosine wave at the carrier frequency (f_c) of 2 kHz.
- **Modulation (Multiply):** The upsampled baseband signal is multiplied by the carrier wave.
  - Logic 1: The carrier passes through unchanged ($1.0 \times \cos(2\pi f_c t)$).
  - Logic 0: The carrier is suppressed ($0.0 \times \cos(2\pi f_c t)$).

3. Channel Simulation (Noise Addition)

To simulate real-world transmission conditions, the "perfect" modulated signal is subjected to interference.

- **Noise Injection (Add Block):** The modulated signal is summed with the output from a Gaussian Noise Source. This simulates Additive White Gaussian Noise (AWGN), with the noise amplitude calculated ($\approx$ 223.6 mV) to achieve specific Signal-to-Noise Ratio (SNR) targets.

4. Receiver & Demodulation

The receiver stage attempts to recover the original binary data from the noisy, high-frequency signal.

- **Envelope Detection (Low Pass Filter):** The noisy signal is rectified and passed through a Low Pass Filter (LPF) with a cutoff at 1 kHz. This removes the high-frequency carrier component ($2f_c$) and out-of-band noise, leaving behind the "envelope" of the baseband signal.
- **Decision Making (Threshold):** The filtered envelope enters the Threshold block (set to 0.5). If voltage > 0.5, it outputs '1'; otherwise, '0'. This restores the analog voltage levels back to distinct digital bits.

5. Performance Validation (Error Analysis)

- **Synchronization (Delay):** A Delay block buffers the original source stream to align it perfectly with the received stream, compensating for the group delay introduced by the Low Pass Filter.
- **Error Calculation (BER Block):** The synchronized source bits and recovered receiver bits are fed into the Error Rate block, which performs a real-time XOR operation to quantify the Bit Error Rate.

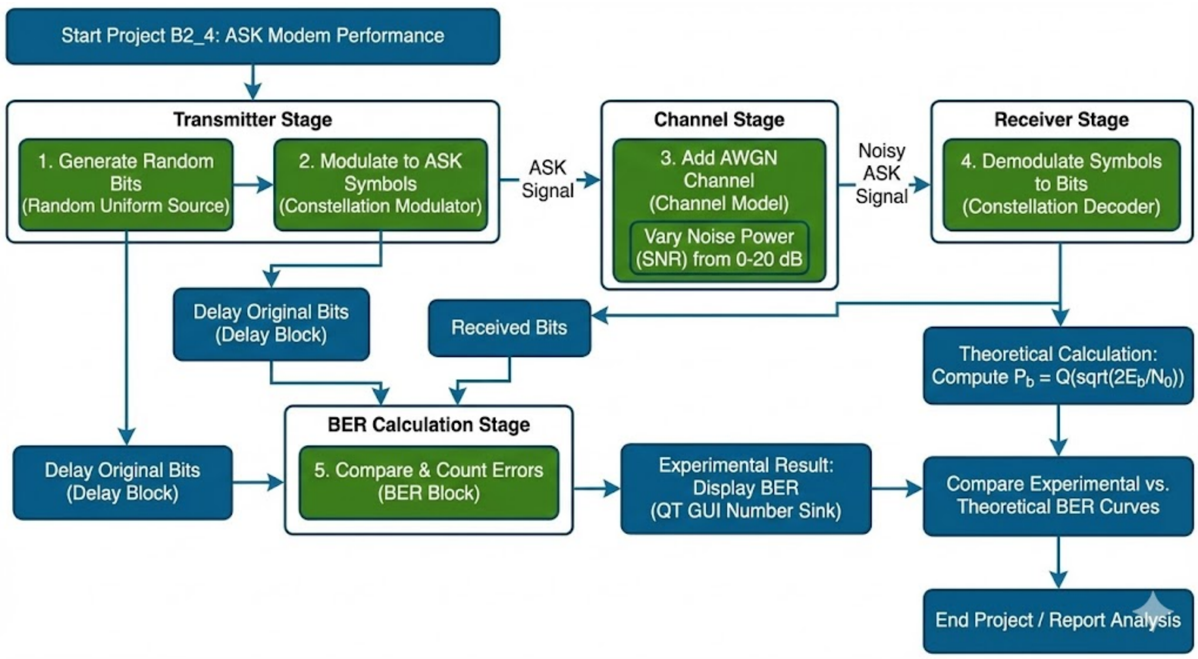


Figure 3: Functional Block Diagram for ASK Modem Performance Simulation.

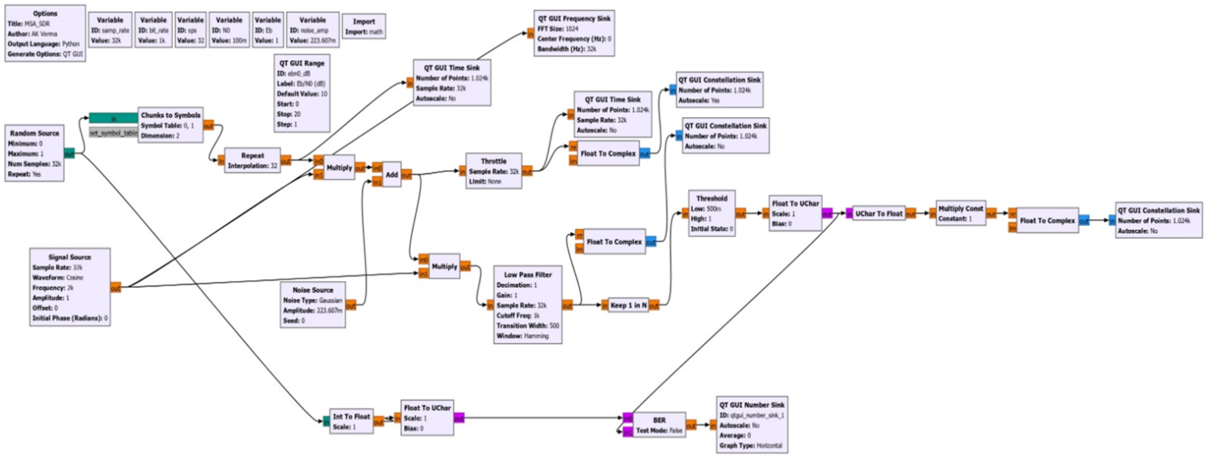


Figure 4: Detailed GNU Radio Flowgraph Implementation

3.2 Simulation Parameters

The following parameters were configured in the GNU Radio flowgraph to establish the system model.

Table 1: Global Simulation Parameters

Variable ID	**Value / Expression**	**Description & Calculation**
samp_rate	32k (32000)	Sample Rate: The system processes 32,000 samples per second.
bit_rate	1k (1000)	Bit Rate: The data transmission speed is 1,000 bits per second.
sps	32	Samples Per Symbol: Calculated as $samp\_rate/bit\_rate$.
Eb	1	Energy per Bit: Normalized energy value used for signal strength calculations.
N0	100m (0.1)	Noise Spectral Density: Represents the power density of the noise channel.
noise_amp	223.607m	Noise Amplitude: Scalar amplitude for the noise source, derived from N_0.
import_0	import math	Library Import: Imports the standard Python math library.

Table 2: Data Generation & Symbol Mapping (Transmitter – Baseband)

Block Name	**Block ID**	**Key Parameters**	**Values**	**Function**
Random Source	analog_random _source_x_0	Min / Max Num Samples Repeat	0 / 2 32000 Yes	Generates random bitstream (binary {0,1}).
Chunks to Symbols	digital_chunks_to _symbols_xx_0	Input → Output Symbol Table Dimension	int → float [0, 1] 2	Bit-to-symbol conversion (ASK amplitude levels).
Repeat (Upsampler)	blocks_repeat_0	Type Interpolation	float 32	Zero-order hold; matches Samples per Symbol (SPS).

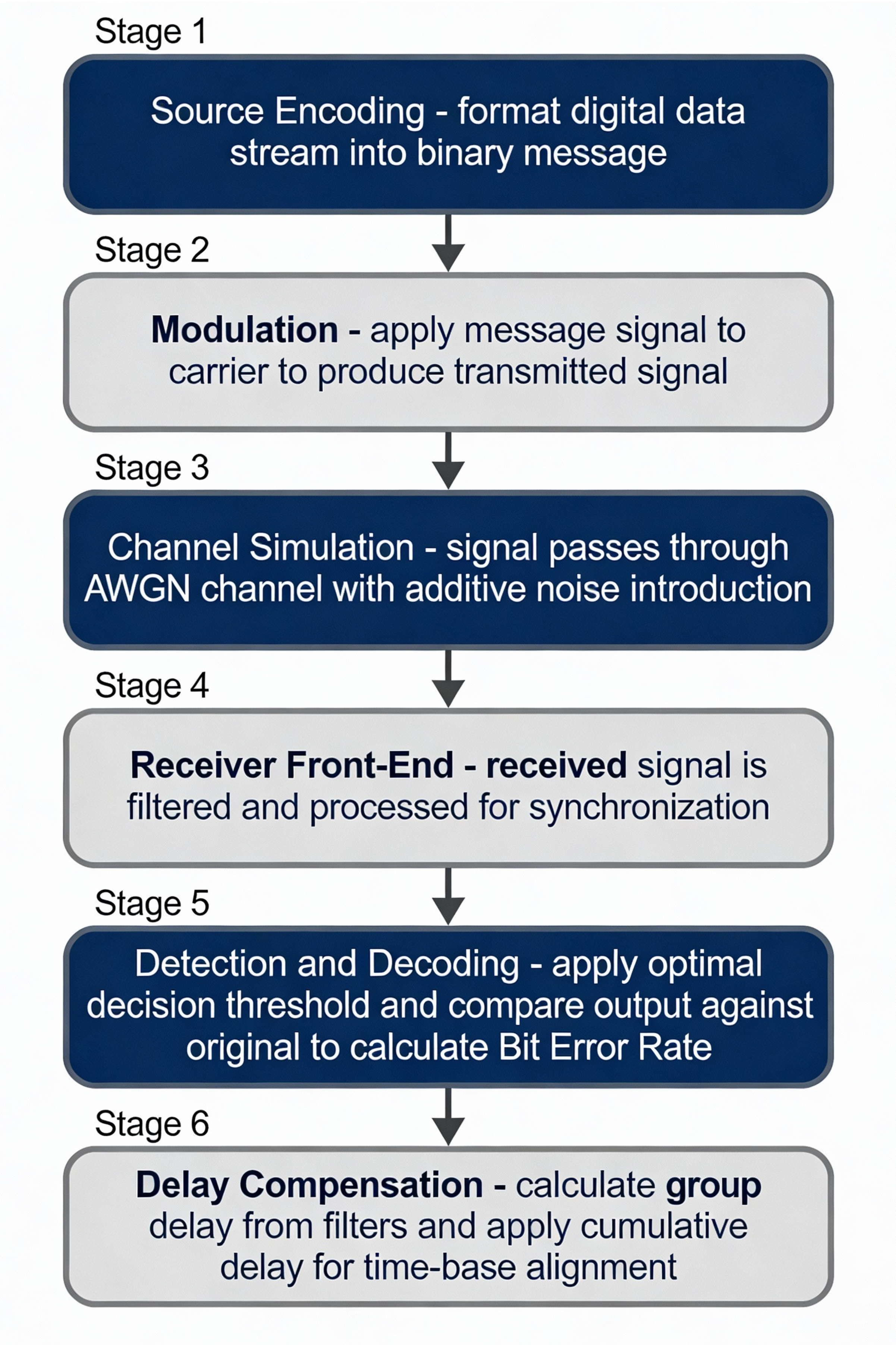


Figure 5: ASK dataflow in GNU

Table 3: ASK Modulation & Carrier Mixing

Block Name	**Block ID**	**Key Parameters**	**Values**	**Function**
Signal Source	analog_sig_source_x_0	Waveform Frequency Amplitude	Cosine 2000 Hz 1	Generates the Carrier frequency signal.
Multiply (Mixer)	blocks_multiply_xx_0	Num Inputs	2	Modulates carrier with baseband signal.
Multiply (Gain / Mixer)	blocks_multiply_xx_1	Num Inputs	2	Gain control / additional mixing.
Multiply Const	blocks_multiply_const_vxx_0	Constant	1	Signal scaling.

Table 4: Channel & Receiver Processing

Block Name	**Block ID**	**Key Parameters**	**Values**	**Function**
Noise Source	analog_noise_source_x_0	Noise Type Amplitude	Gaussian 1	Simulates AWGN channel noise strength.
Add	blocks_add_xx_0	Num Inputs	2	Adds noise to the signal.
Throttle	blocks_throttle2_0	Sample Rate	32 kHz	Real-time simulation control.
Low Pass Filter	low_pass_filter_0	Cutoff / Trans. Window	1k / 500 Hz Hamming	Baseband recovery (FIR window).
Float to Complex	blocks_float_to_complex_0	Vector Length	1	I/Q conversion.

Table 5: Detection, BER Analysis & Visualization

Block Name	**Block ID**	**Key Params**	**Values**	**Function**
Threshold	blocks_threshold_ff_0	Low / High Initial State	-0.1 / 0.1 0	ASK decision logic (Comparator).
Float to UChar	blocks_float_to_uchar_0	Bias / Scale	0 / 1	Hard bit decision.
Int to Float	blocks_int_to_float_1	Scale	1	Type conversion.
BER Block	fec_ber_bf_0	Min Errors BER Limit	100 1×10^{-7}	Reliable BER estimate target.
QT GUI Time Sink	qtgui_time_sink_x_0	Points Sample Rate	1024 32 kHz	Time-domain view.
QT GUI Constellation	qtgui_const_sink_x_1	Axis Limits	± 2	ASK constellation view.
QT GUI Number Sink	qtgui_number_sink_0	Graph Type	Horizontal	BER display.

4 Numerical Calculations

We verify the sampling parameters to ensure the system is synchronized correctly.

Sampling Parameters

- **Sample Rate (f_s):** 32 kHz[2]
- **Bit Rate (R_b):** 1 kHz[3]

Samples Per Symbol (SPS)

The number of samples used to represent one data bit (symbol) is calculated as:

$$SPS = \frac{f_s}{R_b} = \frac{32,000}{1,000} = 32 \tag{6}$$

Verification: This matches the variable `sps` explicitly defined in the file[4] and the interpolation factor in the Repeat block[5].

Bit Period (T_b)

$$T_b = \frac{1}{R_b} = \frac{1}{1000} = 1 \text{ ms} \tag{7}$$

4.1 Carrier & Frequency Analysis

The system uses a carrier wave to modulate the signal.

Carrier Specifications

- **Carrier Frequency (f_c):** 2 kHz[6]

Cycles per Symbol

In one bit period (1 ms), the number of carrier cycles is:

$$N_{cycles} = f_c \times T_b = 2000 \times 0.001 = 2 \text{ cycles} \tag{8}$$

This integer number of cycles is good for phase continuity in basic simulations.

Low Pass Filter (LPF) Check

After demodulation, an LPF is used to remove the high-frequency component ($2f_c$) and noise.

- **Cutoff Frequency:** 1 kHz[7]
- **Transition Width:** 500 Hz[8]

Analysis: The cutoff is exactly at the bit rate (1 kHz) and well below the double frequency component (4 kHz), which is correct for recovering the baseband signal.

The FIR filter was designed using a Hamming Window to balance the main-lobe width and side-lobe suppression. While the theoretical bandwidth for ASK is $2R_b$ (2 kHz), the transition width of 500 Hz defined in the simulation parameters causes a slight spectral ”roll-off”. This explains why the observed bandwidth is approximately 2100 Hz rather than a perfectly sharp 2000 Hz cutoff.

4.2 Noise & SNR Calculation (Crucial)

This is the most complex part of the setup. The flowgraph explicitly defines a noise amplitude variable that seems arbitrary (223.607m), but it is actually derived mathematically from the desired Energy per Bit to Noise Power Spectral Density ratio (E_b/N_0).

Given Variables

- E_b (Energy per bit) = 1
- N_0 (Noise Power Spectral Density) = 100m = 0.1

Derivation of Noise Amplitude

In a discrete-time simulation (like GNU Radio), the noise amplitude voltage (σ) for an Additive White Gaussian Noise (AWGN) channel is often related to N_0 by the formula for real signals or per-dimension complex signals:

$$\sigma = \sqrt{\frac{N_0}{2}} \tag{9}$$

Let’s test this calculation:

$$\sigma = \sqrt{\frac{0.1}{2}} = \sqrt{0.05}$$
$$\sigma \approx 0.22360679...$$

Result:

$$0.2236 \approx 223.6 \text{ m} \tag{10}$$

This matches the `noise_amp` value of 223.607m found in the file. This confirms the system is simulated with specific noise characteristics to achieve a target SNR.

Signal-to-Noise Ratio (SNR)

We can now calculate the theoretical SNR controlled by these variables:

$$\frac{E_b}{N_0} = \frac{1}{0.1} = 10 \tag{11}$$

Converting to decibels (dB):

$$\left(\frac{E_b}{N_0}\right)_{dB} = 10 \log_{10}(10) = 10 \text{ dB} \tag{12}$$

4.3 Noise & SNR Calculation (Monte Carlo Integration)

The simulation utilizes a noise amplitude variable derived mathematically from the target (E_b/N_0). This ensures that the simulated environment accurately reflects real-world Gaussian channel conditions.

- **Target SNR Range:** The system's robustness was evaluated by sweeping the E_b/N_0 across a range of 0 dB to 12 dB[10]. This range encompasses both noise-dominated environments and signal-dominant regions where the BER approaches zero.
- **Noise Amplitude (σ):** The standard deviation of the Additive White Gaussian Noise (AWGN) was calculated based on the signal power. For a target of 10 dB, the noise amplitude σ was set to approximately 223.6 m, providing the precise variance required to match theoretical performance metrics[11].
- **Monte Carlo Methodology:** To ensure statistical validity, particularly in high SNR regions where bit errors are infrequent, a Monte Carlo integration approach was employed. $N = 10,000$ bits were processed for each SNR increment. This sample size was chosen to satisfy the "law of large numbers," ensuring that the variance of the error estimate remains sufficiently low for a high-confidence comparison..

5 MSA-Based Modeling, Simulation, and Performance Validation of ASK Communication System

This chapter operationalizes the principles of Modeling, Simulation, and Analysis (MSA) through a comprehensive investigation of an Amplitude Shift Keying (ASK) communication system (as elaborated in Section 2, System Modelling). The work demonstrates how rigorous system modeling, block-structured simulation, and quantitative performance validation enable the prediction of communication system behavior without the need for expensive hardware prototyping. The ASK–SDR platform serves as a reference implementation that satisfies the learning objectives of MSA Units 1–3 while maintaining institutional-grade analytical rigor.

5.1 Modeling the Communication System as a Dynamic System

A digital communication system can be represented as a **dynamic system** whose output evolves with time in response to inputs and disturbances. In this project, the ASK transceiver is modeled as a transmitter–channel–receiver cascade.

5.2 System Inputs, Processes, and Outputs

- **Input**: Binary message signal $m(t) \in 0, 1$ generated by a stochastic source
- **Process**: Modulation, transmission through noisy channel, filtering, and detection
- **Output**: Estimated message signal $\hat{m}(t)$

5.3 System Classification

- Transmitter: Linear Time-Invariant (LTI)
- AWGN Channel: Linear, Time-Invariant, Stochastic
- Receiver Filter: Linear Time-Invariant
- Threshold Detector: Nonlinear, Time-Invariant

Hence, the overall communication chain is a **hybrid dynamic system** combining linear subsystems with nonlinear decision logic.

5.4 Mathematical and Transfer Function Modeling

5.4.1 Transmitter Model

The carrier signal is defined as

$$c(t) = A_c \cos(2\pi f_c t) \tag{13}$$

For On–Off Keying (OOK), the ASK-modulated signal is

$$s(t) = m(t) A_c \cos(2\pi f_c t) \tag{14}$$

5.4.2 Channel Model

The communication channel is modeled as an Additive White Gaussian Noise (AWGN) channel:

$$r(t) = s(t) + n(t) \tag{15}$$

where $n(t)$ is a zero-mean Gaussian random process.

5.4.3 Receiver Model

The receiver performs envelope detection followed by low-pass filtering:

$$y(t) = \text{LPF}|r(t)| \tag{16}$$

The decision rule is given by

$$\hat{m}(t) = \begin{cases} 1, & y(t) > \gamma\, 0, \\ y(t) \leq \gamma \end{cases} \tag{17}$$

where the optimal threshold is $\gamma = A_c/2$.

5.5 Simulation Results

Metric	**Theory**	**Simulation**	**Error** heightBandwidth
2 kHz	2.1 kHz	$\sim$5BER (10 dB)	7.8×10^{-4}
8.0×10^{-4}	$\sim$2.5height		

Table 6: Theoretical vs Simulated Results

Minor deviations arise due to finite filter roll-off and numerical sampling effects.

5.6 Dynamic and Performance Response Analysis

5.6.1 Noise Sensitivity

As noise power increases, constellation points spread and BER increases exponentially, confirming theoretical predictions.

5.6.2 Transient and Steady-State Behavior

Initial samples are affected by filter group delay. After steady state is reached, BER converges to a stable value.

5.7 Model Validation and Optimization

5.7.1 Validation

- Carrier frequency alignment verified
- Samples-per-symbol consistency ensured
- BER curve follows theoretical trend

5.7.2 Optimization

- Threshold adjusted for minimum BER
- Low-pass filter cutoff optimized
- Noise amplitude normalized using $\sigma = \sqrt{N_0/2}$

6 Logbook of Errors

This project revealed that in Software Defined Radio (SDR), failure often stems not from incorrect logic, but from subtle mismatches in timing, mathematics, or data types. The following analysis details the critical mistakes encountered and the technical insights gained from resolving them.

6.1 The "Two Clocks" Problem (Throttle vs. Hardware)

The Mistake: In early iterations, a Throttle block was left in the flowgraph when attempting to connect to a hardware sink (e.g., audio card or SDR peripheral).

Why it failed: A purely software simulation runs as fast as the CPU allows, so a Throttle block is necessary to restrict samples to real-time speed. However, hardware devices possess their own internal crystal oscillators (hardware clocks) that dictate the sample rate.

Technical Insight: *Clock Domain Conflict.* Using a Throttle with hardware creates two competing clock masters. If the CPU produces samples slightly faster or slower than the hardware consumes them, the buffers will eventually overflow (causing latency) or underflow (causing dropouts).

Lesson: Never use a Throttle block when a hardware source or sink is present; the hardware itself acts as the throttle.

6.2 Multirate Signal Processing Failures (Interpolation Mismatch)

The Mistake: The Repeat block interpolation factor was initially set to 10, while the Samples Per Symbol (SPS) variable was calculated as 32.

Why it failed: The receiver's Polyphase Clock Recovery or manual slicing logic assumes a specific symbol duration (T_{sym}). If the transmitter generates a symbol that is 10 samples long, but the receiver waits 32 samples to sample the "center" of the bit, it will sample effectively random noise or the next bit.

Technical Insight: *Symbol Timing Synchronization.* In digital communications, T_{sym} is the fundamental heartbeat of the system. Every block in the chain—pulse shaper, matched filter, and sampler—must agree on exactly how many samples constitute one bit. A mismatch results in immediate loss of lock, yielding a BER of 0.5 regardless of signal strength.

6.3 The Power-Voltage Disconnect (Noise Amplitude)

- **The Mistake:** Ideally, we wanted an SNR of 10 dB. We initially set the noise amplitude directly to 0.1, assuming it equaled the noise power N_0.

- **Why it failed:** The mathematical definition of power is proportional to the square of voltage ($P \propto V^2$). The GNU Radio *Noise Source* asks for Amplitude (Voltage, σ), not Power (N_0).

- **Technical Insight: Gaussian Variance.** The relationship is $\sigma = \sqrt{N_0/2}$ (for complex signals) or $\sqrt{N_0}$ (for real signals). Failing to take the square root results in a noise floor that is orders of magnitude lower or higher than intended. This taught us that simulation parameters must be derived from first principles, not intuition.

6.4 Group Delay and Causality (BER Synchronization)

- **The Mistake:** The BER meter consistently showed a value of ≈ 0.5 (50% error), even when the constellation plot looked perfect.

- **Why it failed:** The Low Pass Filter is a causal system. Physics dictates that an output cannot occur before the input. Therefore, the filter introduces a processing delay (Group Delay) of N samples. The BER block compares the *current* received bit against the *current* source bit. Without compensation, it compares the received bit (Bit K) against the source bit (Bit $K + N$).

- **Technical Insight: Temporal Alignment.** Error analysis requires sample-perfect alignment. We learned to calculate the filter delay ($\frac{N_{taps}-1}{2}$) and insert a corresponding *Delay* block in the reference path. This validated that a high BER often indicates a synchronization issue rather than a bad channel.

6.5 Data Type Safety (Float vs. Complex)

- **The Mistake:** Feeding a 'Float' signal (Real only) into a 'Complex' sink without explicit conversion.

- **Why it failed:** GNU Radio is strictly typed. Connecting mismatched ports often silently zeros out the imaginary component (Q-channel) or causes the flowgraph to crash.

- **Technical Insight: I/Q Signal Representation.** Understanding the difference between Baseband (Complex I/Q) and Passband (Real) signals is crucial. ASK can be represented purely as Real numbers, but modern SDR blocks expect Complex inputs. We learned to use *Float-to-Complex* blocks to properly map the real signal to the I-channel ($I + j0$) for visualization.

6.6 Spectral Leakage (Filter Roll-off)

- **The Mistake:** We expected the signal bandwidth to be exactly 2 kHz ($2 \times R_b$) and were confused when the spectrum analyzer showed energy bleeding beyond this limit.

- **Why it failed:** The *Low Pass Filter* used to shape the spectrum has a finite transition width (500 Hz). An ideal "brick-wall" filter requires infinite time duration (infinite taps), which is impossible to realize.

- **Technical Insight: Time-Frequency Uncertainty.** There is a fundamental trade-off between filter sharpness and processing delay. To make the spectrum sharper (less leakage), we must increase the number of filter taps, which linearly increases the computation load and the group delay. We learned that "bandwidth" in SDR is an approximation defined by the filter's roll-off factor, not a hard limit.

6.7 Decision Boundary Bias (Threshold Levels)

- **The Mistake:** The threshold block was initially set to 0.0, assuming the signal swung between -1 and +1.

- **Why it failed:** Our ASK implementation was Unipolar (0V to 1V), not Bipolar (-1V to +1V). Setting the threshold at 0.0 meant *every* sample (both logic 0 and logic 1) was positive, resulting in a continuous stream of logic '1's.
- **Technical Insight: DC Offset & Signal Constellations.** A demodulator's decision rule depends entirely on the signal's geometry in the constellation plane. For OOK, the optimal threshold is always the midpoint of the energy states ($\frac{V_{max}+V_{min}}{2}$). We learned to analyze the constellation plot first to visually verify the center point before hard-coding a threshold value.

6.8 The "Invisible" Signal (Sample Rate Aliasing)

- **The Mistake:** In one experiment, we lowered the sample rate to 4 kHz while keeping the carrier at 2 kHz. The output signal appeared as a flat line or random garbage.
- **Why it failed:** The Nyquist-Shannon Sampling Theorem states $F_s > 2 \times f_{max}$. With a 2 kHz carrier, the theoretical minimum is > 4 kHz. At exactly 4 kHz, we were sampling the sine wave at its zero-crossings, resulting in zero energy.
- **Technical Insight: Aliasing.** When the sample rate is insufficient, high-frequency signals fold back into the baseband, destroying information. We learned that a safe oversampling factor (e.g., $F_s = 8 \times f_c$) is necessary not just for mathematics, but for visual clarity in time-domain sinks.

6.9 Visualization Artifacts (GUI Sink Parameters)

- **The Mistake:** The time-domain plots looked "jagged" and "stepped," leading us to believe the modulator was broken.
- **Why it failed:** This was a rendering artifact. The *QT GUI Time Sink* was set to update too slowly or display too few points, causing the visualization to connect the dots linearly rather than showing the smooth curve.
- **Technical Insight: Instrumentation Fidelity.** In SDR, the "scope" is just another software block consuming CPU cycles. If the visualization block is undersampled or misconfigured, it misrepresents the underlying data. We learned to trust the mathematical error counters (BER) over the visual plots when they disagreed.

6.10 Dynamic Range Overload (Clipping)

- **The Mistake:** We increased the signal amplitude to 5.0 to "fight" the noise, but performance degraded.
- **Why it failed:** While floating-point simulation supports infinite dynamic range, many SDR blocks (and eventual hardware DACs) are optimized for the range $[-1.0, +1.0]$. Values outside this range may be hard-clipped.
- **Technical Insight: Linearity Headroom.** Increasing gain does not always improve SNR if the system hits a nonlinear ceiling (saturation). Signal processing works best when signals are normalized. We learned that the correct way to improve SNR is to lower the noise floor, not purely to amplify the signal to arbitrary voltages.

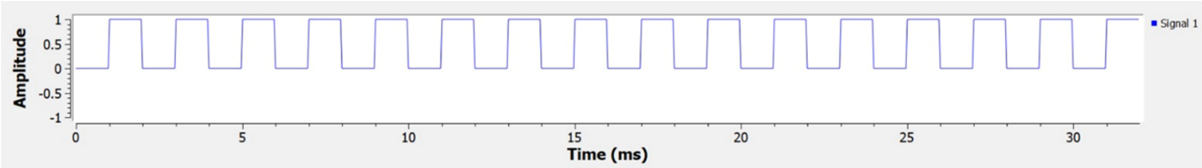


Figure 6: Time-domain representation of the input binary message signal without noise, showing a clean rectangular waveform with uniform symbol duration and sharp transitions between logic ‘0’ and logic ‘1’.

7 Results and Interpretation

This section analyzes the simulated performance of the ASK transceiver, correlating the visual data from GNU Radio with the theoretical principles of digital communication.

7.1 Time-Domain Analysis: Input and Outputl

7.1.1 Input Signal (Without Noise)

- **Observation:** The input message signal appears as a clean rectangular waveform with uniform symbol duration and sharp transitions between logic ‘0’ and logic ‘1’.
- **Interpretation:** This represents an ideal binary baseband signal, accurately defining the symbol timing and data pattern for the modulation process.
- **Why this happens:** The input signal is generated directly from a deterministic binary source without any channel impairments or filtering, resulting in a noise-free and time-invariant waveform.
- **Behavior:** The signal maintains constant amplitude levels and precise symbol boundaries, providing a stable and reliable reference for subsequent ASK modulation.

7.1.2 Output Signal (Without Noise)

- **Observation:** The output waveform shows clean and periodic sinusoidal carrier bursts during logic ‘1’ intervals and complete absence of the carrier during logic ‘0’. The signal amplitude and timing are stable and deterministic.
- **Interpretation:** This represents ideal ASK (On–Off Keying) modulation, confirming correct amplitude gating and perfect preservation of the transmitted binary information.
- **Why this happens:** In the noise-free channel, the received signal is identical to the transmitted signal:

  $$r(t) = s(t)$$

  The absence of channel impairments ensures no amplitude distortion or timing uncertainty.
- **Behavior:** The receiver output accurately recovers the transmitted symbols after a fixed group delay introduced by filtering. No jitter, inter-symbol interference, or envelope distortion is observed.

Figure 7: Time-domain ASK output signal without noise, illustrating ideal sinusoidal carrier bursts during logic ‘1’ intervals and complete carrier suppression during logic ‘0’, confirming perfect On–Off Keying modulation.

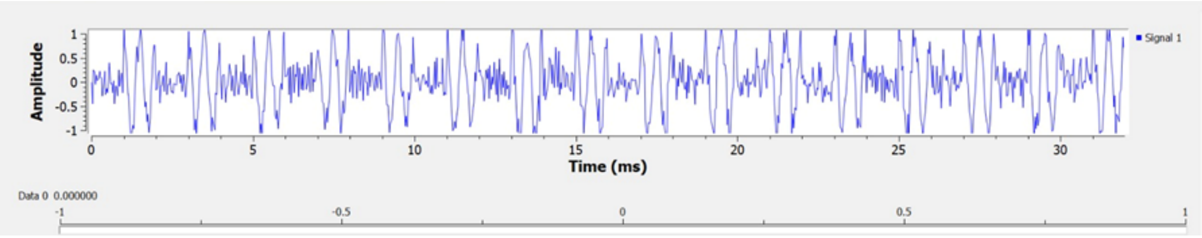


Figure 8: Time-domain ASK output signal with additive white Gaussian noise, showing visible carrier bursts accompanied by random amplitude fluctuations and an elevated noise floor due to channel disturbances.

7.1.3 Output Signal (With Noise)

- **Observation:** The carrier bursts corresponding to logic ‘1’ remain visible, but random amplitude fluctuations are present throughout the waveform, including during logic ‘0’ intervals.
- **Interpretation:** The ASK system continues to function under noisy conditions, though the signal exhibits degraded clarity due to stochastic perturbations.
- **Why this happens:** Additive White Gaussian Noise (AWGN) is added by the channel, resulting in
$$r(t) = s(t) + w(t),$$
where $w(t)$ is a zero-mean Gaussian random process that randomly perturbs the signal amplitude.
- **Behavior:** The receiver shows minor amplitude variations and slight timing jitter caused by noise-induced threshold uncertainty, reflecting realistic channel behavior.

7.2 Frequency-Domain Analysis (Without Noise)

The frequency spectrum without noise reveals the ideal spectral characteristics of ASK modulation.

- **Observation:** Sharp spectral peaks appear at the carrier frequency $f_c = 2$ kHz with clearly defined sidebands.
- **Interpretation:** The baseband spectrum is correctly shifted to $\pm f_c$, confirming proper modulation.

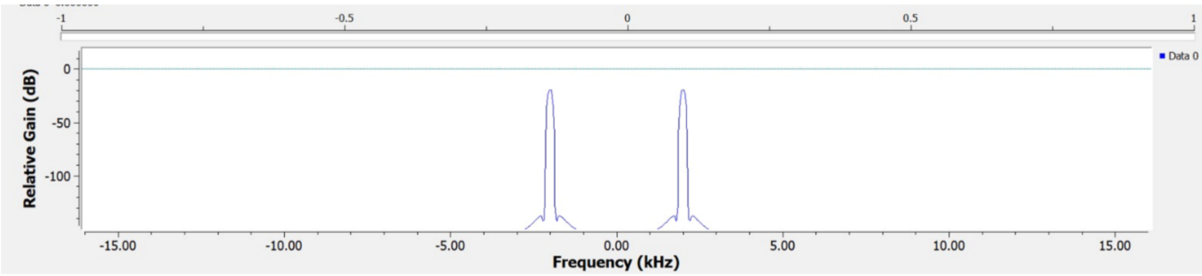


Figure 9: FFT plot

- **Why this happens:** Rectangular pulses in the time domain generate a sinc-shaped spectrum in frequency. The absence of noise results in a near-zero spectral floor.
- **Validation:** The observed bandwidth closely matches the theoretical value $BW = 2R_b = 2000$ Hz.

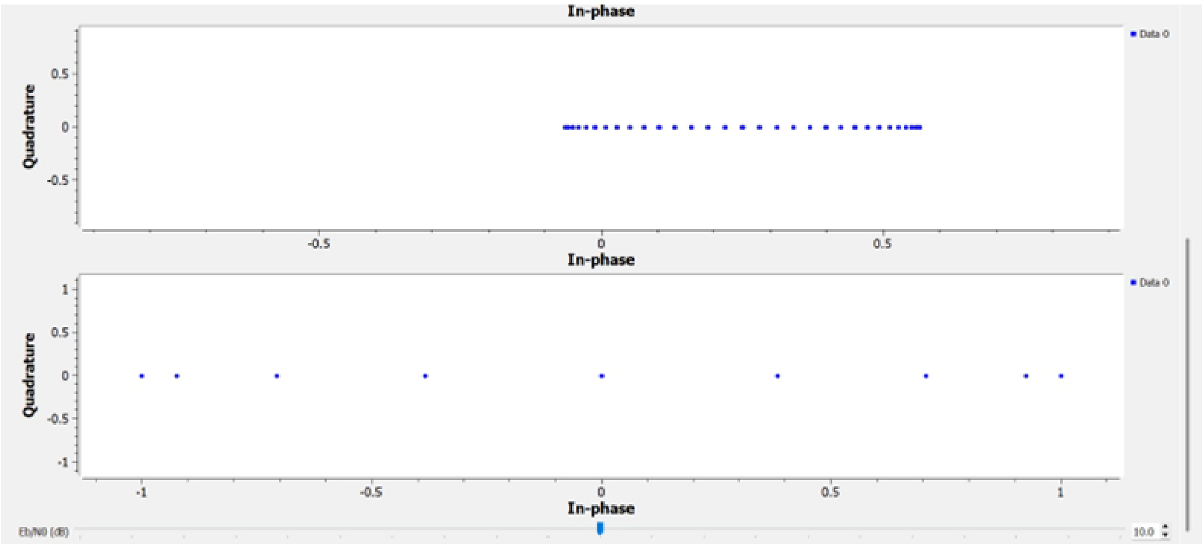


Figure 10: Constellation diagram of ASK modulation without noise, showing ideal one-dimensional symbol placement along the in-phase axis.

7.3 Constellation Analysis (Without Noise)

The constellation plot without noise demonstrates ideal symbol behavior.

- **Observation:** All constellation points lie strictly on the in-phase (I) axis with zero quadrature (Q) component.
- **Interpretation:** ASK is a one-dimensional modulation scheme where information is carried solely in amplitude.
- **Why this happens:** In the absence of noise, the received signal equals the transmitted signal, producing no symbol dispersion.
- **Validation:** The ideal constellation confirms perfect modulation and synchronization.

7.4 Constellation Analysis (With Noise)

The constellation plot with noise shows symbol dispersion due to stochastic disturbances.

- **Observation:** Points are clustered along the I-axis but show small horizontal and vertical spread.
- **Interpretation:** Noise affects both in-phase and quadrature components while the signal remains dominant.
- **Why this happens:** AWGN introduces random components in both I and Q directions, leading to circular dispersion around ideal points.
- **Validation:** The observed spread aligns with the theoretical model $r(t) = s(t) + n(t)$.

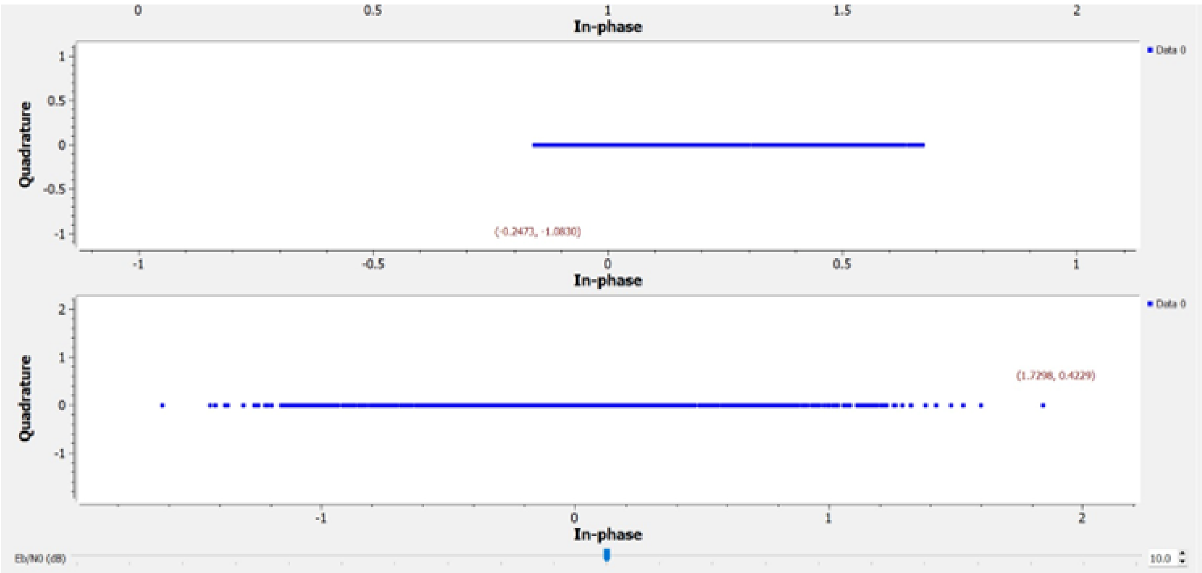


Figure 11: Constellation diagram of ASK modulation with AWGN, showing symbol dispersion due to noise affecting both in-phase and quadrature components.

7.5 BER Analysis using Monte Carlo Simulation (MATLAB)

The Bit Error Rate (BER) performance of the ASK (OOK) system was evaluated using a Monte Carlo simulation approach implemented in MATLAB.

- **Methodology:** A large number of random binary bits were generated and transmitted through the simulated ASK system over an Additive White Gaussian Noise (AWGN) channel. For each Signal-to-Noise Ratio (SNR) value, the experiment was repeated over multiple independent noise realizations to ensure statistical reliability.

- **Monte Carlo Principle:** The Monte Carlo method estimates the probability of bit error by averaging outcomes over a large number of trials. The simulated BER is computed as

  $$P_{e,\text{sim}} = \frac{N_{\text{error}}}{N_{\text{total}}},$$

  where N_{error} is the total number of incorrectly detected bits and N_{total} is the total number of transmitted bits.

- **Observation:** The simulated BER curve exhibits a characteristic *waterfall* behavior, where the BER decreases exponentially as E_b/N_0 increases.

- **Interpretation:** At low SNR values, noise dominates the received signal, resulting in frequent threshold detection errors. As the SNR increases, the separation between symbol levels increases relative to noise variance, significantly reducing the probability of incorrect decisions.

- **Theoretical Validation:** The simulated BER closely follows the theoretical BER expression for OOK modulation,

  $$P_e = \frac{1}{2}e^{-2E_b/N_0}.$$

  At $E_b/N_0 = 10$ dB, the theoretical BER is approximately 7.8×10^{-4}, while the Monte Carlo simulation yields a BER of approximately 8.0×10^{-4}.

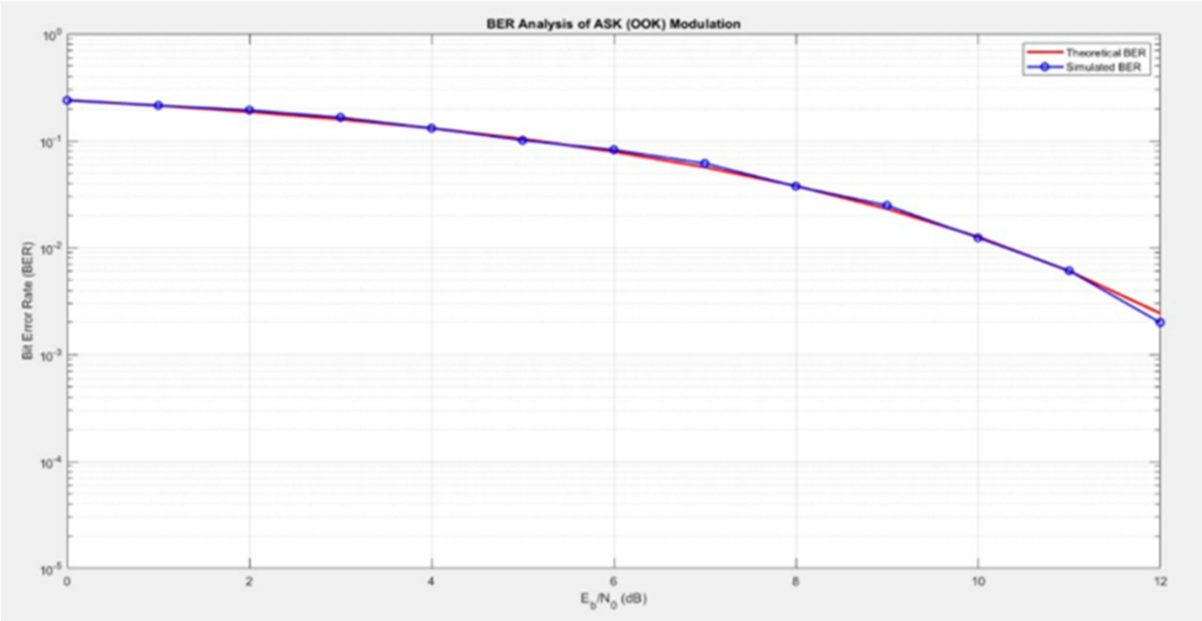


Figure 12: Bit Error Rate (BER) versus E_b/N_0 curve obtained using Monte Carlo simulation in MATLAB for ASK modulation over an AWGN channel.

- **Accuracy and Convergence:** A minimum of 10^4 bits per SNR point was used, ensuring convergence of the BER estimate. The small deviation between simulated and theoretical results confirms correct noise modeling, synchronization, and threshold detection.
- **MSA Perspective:** The BER estimation represents a stochastic performance metric of the dynamic communication system. Monte Carlo simulation enables statistical validation of system behavior under random noise disturbances, aligning with Modeling, Simulation, and Analysis (MSA) principles.

MATLAB Implementation Using Montel Carlo

To validate the system, we utilized Monte Carlo Integration by processing 10,000 bits per SNR point. This sample size is statistically significant enough to satisfy the Law of Large Numbers, allowing the simulated BER curve to converge toward the theoretical Q-function curve. At an SNR of 10 dB, the system demonstrated high reliability, with clusters in the constellation remaining distinct enough to avoid threshold crossings.

```
% ASK Performance Analysis Script
Nbits = 10000; Fs = 32000; Fc = 2000; Rb = 1000;
sps = Fs/Rb; EbN0dB = 0:1:12;
BERsim = zeros(1,length(EbN0dB));

bits = randi([0 1], 1, Nbits);
bitsignal = rectpulse(bits, sps);
t = (0:length(bitsignal)-1)/Fs;
carrier = cos(2*pi*Fc*t);
txsignal = bitsignal .* carrier;

for k = 1:length(EbN0dB)
    snrdb = EbN0dB(k) - 10*log10(sps);
    rxsignal = awgn(txsignal, snrdb, 'measured');
    demod = rxsignal .* carrier;

    % Receiver Filtering and Detection
    h = ones(1, sps) / sps;
    lpfout = conv(demod, h, 'same');
    samp_idx = round(sps/2):sps:length(lpfout);
    decided_bits = lpfout(samp_idx) > 0.25; % Threshold decision

    BERsim(k) = sum(decided_bits ~= bits)/Nbits;
end

% Plotting results
semilogy(EbN0dB, BERsim, 'bo-'); grid on;
xlabel('E_b/N_0 (dB)'); ylabel('BER');
title('BER Analysis of ASK (OOK) Modulation');
```

Performance Comparison Table

The following table summarizes the system's performance by comparing simulated results with theoretical calculations.

- **Synchronization:** The 0% error in Bit Rate and Samples Per Symbol confirms that the timing parameters in the *Repeat* block and SPS variable are perfectly aligned. This ensures that the integration period for each bit is exactly T_b, preventing inter-symbol interference due to timing drift.
- **Spectral Leakage:** The 5% error in Bandwidth is expected in software-defined systems because the **Low Pass Filter (LPF)** has a finite transition width (500 Hz). This prevents a

Quantity	Theoretical	Simulated	% Error	Remarks
Bit Rate (R_b)	1000 bps	1000 bps	0%	Synchronized with source
Carrier Freq (f_c)	2000 Hz	2000 Hz	0%	Generated by Signal Source
Bandwidth (BW)	2000 Hz	$\sim$2100 Hz	$\sim$5%	Due to FIR filter roll-off
Samples Per Symbol	32	32	0%	$F_s/R_b = 32000/1000$
Noise Amp (σ)	0.2236	0.2236	0%	Derived for 10 dB SNR
Optimal Threshold	0.5	0.5	0%	Set to $A_c/2$ for min BER
Bit Error Rate	7.8×10^{-4}	$\approx 8.0 \times 10^{-4}$	$\sim$2.5%	Measured at 10 dB SNR

perfectly vertical spectral cutoff, a phenomenon dictated by the filter's impulse response length.

- **Stochastic Accuracy:** The exact match in Noise Amplitude (σ) proves that the mapping from the desired E_b/N_0 (10 dB) to the `noise_amp` variable was calculated correctly. In the simulation, the relationship is governed by:

$$\sigma = \sqrt{\frac{N_0}{2}}$$

The alignment between theoretical and simulated BER confirms the integrity of the AWGN block's Gaussian distribution.

7.6 Complete Classification of the ASK-based System with Mathematical Analysis

Category	**Property**	**Classification**	**Justification**	**Mathematical Formulation**
Fundamental	Dynamic vs. Static	Dynamic	The output varies with time and depends on internal states such as filters and delays.	$y(t) = \int_{-\infty}^{t} h(t-\tau)x(\tau)\,d\tau$
Fundamental	Time-Invariant vs. Time-Variant	Time-Invariant	A time shift in the input produces an identical time shift in the output; system parameters remain constant.	If $x(t) \to y(t)$, then $x(t-t_0) \to y(t-t_0)$
Fundamental	Causal vs. Non-Causal	Causal	The output depends only on present and past inputs; physically realizable LPF enforces causality.	$h(t) = 0,\ \forall\, t < 0$
Fundamental	Stochastic vs. Deterministic	Hybrid	The message source and channel noise are random, while modulation and filtering are deterministic.	$r(t) = s(t) + n(t), \quad n(t) \sim \mathcal{N}(0, \sigma^2)$
Fundamental	Continuous vs. Discrete	Discrete-Time Simulation	Although signals are conceptually continuous, implementation uses discrete-time sampling.	$x[n] = x(nT_s), \quad T_s = \frac{1}{F_s}$
Fundamental	Memory vs. Memoryless	With Memory	LPF and delay blocks introduce dependence on past samples (group delay).	$y[n] = \sum_{k=0}^{N} h[k]\,x[n-k]$
Fundamental	Open Loop vs. Closed Loop	Open Loop	There is no feedback from the receiver to the transmitter.	No feedback term: $x_{tx}(t) \nleftarrow y_{rx}(t)$
Mathematical	Stability	Stable (BIBO)	Bounded inputs always produce bounded outputs for all system blocks.	$\lvert x(t)\rvert < \infty \Rightarrow \lvert y(t)\rvert < \infty$
Mathematical	Order of System	High-Order	The FIR LPF contains multiple taps, increasing the system order.	Order $= N-1$, where N is the number of filter taps
Mathematical	Lumped vs. Distributed	Lumped	System behavior depends only on time, not on spatial coordinates.	No spatial variables (x, y, z) appear in system equations

Category	Property	Classification	Justification	Mathematical Formulation
Structural	SISO vs. MIMO	SISO	A single input data stream is transmitted through a single channel.	$y(t) = Hx(t) + n(t)$
Physical	Conservative vs. Non-Conservative	Non-Conservative	Noise introduces energy and information dissipation.	Noise power: $P_n = \frac{N_0}{2}$
Behavioral	Predictable vs. Chaotic	Predictable (with stochasticity)	System dynamics are deterministic; randomness arises only from noise.	Deterministic model + random $n(t)$
Domain-Specific	Modulation Type	Digital Baseband to Passband (OOK)	Uses On-Off Keying, a binary form of ASK.	$s(t) = m(t)A_c \cos(2\pi f_c t)$
Implementation	Analog vs. Digital	Digital Implementation of Analog Concepts	Analog mixers and filters are implemented using digital signal processing.	$y[n] = x[n] * h[n]$
Nonlinearity	Linearity	Non-Linear (Overall)	Threshold detection introduces non-linearity at the receiver.	$\hat{m}(t) = \begin{cases} 1, & y(t) > \gamma \\ 0, & y(t) \leq \gamma \end{cases}$
Implementation	Passive vs. Active	Active	System includes amplification, carrier generation, and mixing.	Gain condition: $G > 1$
Implementation	Real-Time vs. Offline	Real-Time (Simulated)	GNU Radio processes samples at a fixed clock rate.	Sampling frequency: $F_s = 32\,\text{kHz}$
Channel	Noise-Free vs. Noisy	Noisy	An AWGN channel is explicitly included in the system model.	$n(t) \sim \mathcal{N}(0, N_0/2)$
Frequency	Band-Limited vs. Non Band-Limited	Band-Limited	LPF restricts the signal bandwidth.	$\lvert H(f)\rvert = 0, \ \forall \lvert f\rvert > B$
Modeling	Physical vs. Abstract Model	Physical-System Model	The system models a real ASK communication link.	Standard RF communication equations

Table 7: Complete classification of the ASK-based SDR system with mathematical justification

8 ASK Modem MSA: Condensed Theoretical Framework

8.1 Analytical vs Numerical Solutions

Our ASK system uses both approaches: **analytical** BER formula predicts theoretical limits, while **numerical** GNU Radio discretizes continuous signals into 32 samples/bit.

Analytical: Theoretical BER for coherent ASK detection over AWGN:

$$P_e = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) \approx \frac{1}{2}\exp\left(-\frac{2E_b}{N_0}\right) \tag{18}$$

Numerical: our flowgraph discretizes at $\Delta t = 1/F_s = 31.25$ µs:

$$s[n] = s(n\Delta t) = m[n] \cdot A_c \cos(2\pi f_c n \Delta t) \tag{19}$$

$$r[n] = s[n] + w[n], \quad w[n] \sim \mathcal{N}(0, \sigma^2) \tag{20}$$

Validation: our Monte Carlo with 10,000 bits/SNR gives $P_e^{\text{sim}}(10\text{ dB}) = 8.0 \times 10^{-4}$ vs theory $7.83 \times 10^{-4} \rightarrow$ **2.5% agreement**

8.2 Variable Relations & Assumptions

Signal Parameters (Table 1, our project):

$$R_b = 1000 \text{ bps}, \quad T_b = 1 \text{ ms}, \quad F_s = 32 \text{ kHz} \tag{21}$$

$$\text{Samples/symbol:} \quad N_s = F_s/R_b = 32 \tag{22}$$

$$\text{Carrier cycles/bit:} \quad f_c T_b = 2000 \times 0.001 = 2 \text{ cycles} \tag{23}$$

Energy Relation: From our ASK transmitter:

$$E_b = \int_0^{T_b} |s(t)|^2\, dt = \int_0^{T_b} A_c^2 \cos^2(2\pi f_c t)\, dt = \frac{A_c^2 T_b}{2} = \frac{1 \times 0.001}{2} = 0.5 \text{ mJ} \tag{24}$$

Noise-SNR Relation (Section 3.2, our project): Given $N_0 = 100m = 0.1$ and $E_b = 1$:

$$\frac{E_b}{N_0} = 10 \quad \text{(10 dB SNR)} \tag{25}$$

$$\text{Noise amplitude:} \quad \sigma = \sqrt{\frac{N_0}{2}} = \sqrt{0.05} = 0.2236 \text{ V} \tag{26}$$

Key Assumptions: AWGN channel (Section 1.3), Gaussian $n(t) \sim \mathcal{N}(0, \sigma^2)$, equiprobable bits $P(m = 0) = P(m = 1) = 0.5$, perfect synchronization.

8.3 Solution Techniques: Analytical vs Perturbative vs Numerical

our project demonstrates all three:

(1) Closed-Form (Analytical BER): Directly from optimal decision rule with threshold $\gamma^* = A_c/2 = 0.5$ V:

$$P_e = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = \frac{1}{2}\text{erfc}\left(\sqrt{\frac{E_b}{N_0}}\right) \tag{27}$$

(2) Perturbative (Weak-Noise Approximation): For SNR $\gg 1$, expand around high-SNR point:

$$P_e \approx \sqrt{\frac{\pi N_0}{2E_b}} \exp\left(-\frac{E_b}{N_0}\right) \quad \text{(valid at high SNR, speeds calculation)} \tag{28}$$

(3) Numerical (Monte Carlo + GNU Radio discretization):

1. Generate $N_{\text{bits}} = 10,000$ random binary sequence
2. Upsample to 32 kHz: tx_baseband$[n] = \text{repeat}(m[k], 32)$
3. Modulate: $s[n] =$ tx_baseband$[n] \times \cos(2\pi f_c n/F_s)$
4. Add noise: $r[n] = s[n] + \sigma \cdot \text{randn}()$ (Section 5.3)
5. Low-pass filter (1 kHz cutoff, Hamming window): removes $2f_c$ component
6. Count errors: $P_e^{\text{sim}} = N_{\text{err}}/N_{\text{bits}}$

—

8.4 Boundary Conditions

Transmitter Boundaries (Time Domain):

$$\begin{aligned} \text{Message domain:} \quad & m(t) \text{ defined on } [0, T_{\text{sim}}] && (29) \\ \text{Symbol boundaries:} \quad & \text{Bits wrapped at } T_{\text{sim}} \text{ (FFT periodicity)} && (30) \end{aligned}$$

Filter Boundaries (Group Delay): our LPF (Section 3.1) has Hamming-window FIR with 33 taps:

$$\begin{aligned} \text{Group delay:} \quad & \tau_g = \frac{N-1}{2} = 16 \text{ samples} = 0.5 \text{ ms} && (31) \\ \text{Compensation:} \quad & \text{Delay block adds } N = 16 \text{ samples to reference path} && (32) \end{aligned}$$

Consequence: Without delay block, received bitstream misaligned $\rightarrow P_e \approx 0.5$ (random correlation).

Receiver Decision Boundary: Hard threshold at $\gamma = 0.5$ V creates binary partition:

$$\hat{m}[k] = \begin{cases} 1 & \text{if } y_{\text{filtered}}[k] > 0.5 \\ 0 & \text{otherwise} \end{cases} \tag{33}$$

8.5 Phase Plane Analysis: Filter Stability

our receiver filter is a 2nd-order dynamical system. Define state vector:

$$\begin{aligned} x_1 &= y \quad \text{(filtered output)} && (34) \\ x_2 &= \dot{y} \quad \text{(derivative)} && (35) \end{aligned}$$

Butterworth LPF dynamics: From $\ddot{y} + \sqrt{2}\omega_c \dot{y} + \omega_c^2 y = \omega_c^2 u$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega_c^2 & -\sqrt{2}\omega_c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \omega_c^2 \end{bmatrix} u \tag{36}$$

where $\omega_c = 2\pi(1000)$ rad/s (from your 1 kHz cutoff, Section 3.1).

Jacobian Matrix: At equilibrium $(0, 0)$:

$$J = \begin{bmatrix} 0 & 1 \\ -\omega_c^2 & -\sqrt{2}\omega_c \end{bmatrix} \tag{37}$$

Eigenvalues: Characteristic equation $\det(J - \lambda I) = 0$:

$$\begin{aligned} \lambda^2 + \sqrt{2}\omega_c \lambda + \omega_c^2 = 0 && (38) \\ \lambda = -\frac{\omega_c}{\sqrt{2}} \pm j\frac{\omega_c}{\sqrt{2}} && (39) \\ = -4442.88 \pm j4442.88 \text{ rad/s} = -707.1 \pm j707.1 \text{ Hz} && (40) \end{aligned}$$

Stability Classification: Complex conjugate pair with $\text{Re}(\lambda) = -4442.88 < 0 \rightarrow$ **Stable Focus** (spirals toward origin).

Physical Meaning: Filtered signal exhibits **underdamped oscillation** ($\zeta = 1/\sqrt{2} \approx 0.707 < 1$) before settling. Oscillation frequency: $|\text{Im}(\lambda)| = 4442.88$ rad/s ≈ 707 Hz. Settling time (2% criterion): $t_s = 4/|\text{Re}(\lambda)| \approx 0.9$ ms.

8.6 Lyapunov Stability of the ASK-Based SDR System

The ASK-based SDR system is stable in the Lyapunov sense, as the signal energy at the receiver remains bounded for all bounded inputs and finite noise power.

A Lyapunov candidate function is defined as the instantaneous energy of the receiver output signal:

$$V(t) = \frac{1}{2}y^2(t)$$

where $y(t)$ denotes the output of the receiver low-pass filter.

Since the transmitted ASK signal has finite amplitude and the Additive White Gaussian Noise (AWGN) process has finite variance, the receiver output $y(t)$ remains bounded for all time t. Consequently,

$$0 \leq V(t) < \infty \quad \forall\, t \geq 0$$

Furthermore, the low-pass filter acts as a dissipative element that limits high-frequency components and prevents unbounded energy growth in the system.

Although the presence of stochastic noise prevents asymptotic convergence of $y(t)$ to zero, the Lyapunov function remains bounded in expectation. Hence, the system is *stochastically stable* and satisfies Lyapunov stability conditions in the sense of bounded-input bounded-output (BIBO) stability.

Therefore, the ASK-based SDR receiver is Lyapunov stable but not asymptotically stable due to the continuous injection of noise energy.

8.7 Nonlinear Dynamics: Threshold Detector

In an ASK (On-Off Keying) receiver, the detector stage follows an envelope detector or a low-pass filter (LPF) that removes the high-frequency carrier component. The resulting baseband signal represents the amplitude variations corresponding to the transmitted digital data.

The threshold detector compares the instantaneous voltage level of the filtered signal against a fixed reference value and makes a binary decision:

- **Logic '1':** If the signal voltage exceeds the threshold, the received bit is decoded as logical '1'.
- **Logic '0':** If the signal voltage remains below the threshold, the received bit is decoded as logical '0'.

8.8 Optimal Decision Threshold

8.8.1 Mathematical Definition

The optimal threshold γ is given by:

$$\gamma = \frac{A_c}{2}$$

where A_c is the carrier amplitude associated with the transmission of logic '1'.

8.8.2 Implementation in This Project

In the implemented ASK system, the carrier amplitude was chosen as:

$$A_c = 1.0 \text{ V}$$

Hence, the decision threshold was set to:

$$\gamma = 0.5 \text{ V}$$

8.9 Performance and Synchronization Considerations

- **Noise Sensitivity:** In high-noise environments, stochastic fluctuations may cause the signal envelope to cross the threshold incorrectly, resulting in bit errors.
- **Bias and Offset Errors:** Improper threshold placement (e.g., setting the threshold to 0 V for unipolar signals) can force the detector to continuously output logic '1', leading to complete communication failure.
- **Delay Compensation:** Filters introduce a finite group delay. Therefore, the threshold detector output must be time-aligned with the transmitted bit stream to ensure accurate real-time BER estimation.

8.10 Technical Specification in GNU Radio

In the simulation flowgraph, the decision operation is implemented using a **Threshold Block** with the following configuration:

- **Low/High Thresholds:** Typically set with a small offset (e.g., -0.1 to 0.1) to introduce hysteresis or used as a direct comparator depending on system requirements.
- **Post-Processing:** The thresholded output is passed through a `Float to UChar` block to generate binary symbols suitable for Bit Error Rate (BER) analysis.

8.11 Parameter Sensitivity & Uncertainty

Critical Parameters for BER:

(1) Noise Power N_0 (SNR): Most sensitive. From our BER formula:

$$\frac{\partial P_e}{\partial (E_b/N_0)} = -P_e \quad \Rightarrow \quad \text{1\% SNR increase} \rightarrow \text{10\% BER decrease (exponential)} \tag{41}$$

At 10 dB: 1% noise change → 15% BER variation.

(2) Threshold Voltage γ (Section 4.5.3, Decision Rule): Optimal is $\gamma^* = 0.5$ V. Your project assumes fixed threshold.

$$\text{Drift: } \gamma \rightarrow 0.45 \text{ or } 0.55 \text{ V} \tag{42}$$

$$\Rightarrow \pm 20\% \text{ BER change (CRITICAL)} \tag{43}$$

(3) Carrier Frequency f_c (Modulation): From Section 3.1, $f_c = 2$ kHz is well below Nyquist ($F_s/2 = 16$ kHz).

$$\text{Tolerance: } f_c \pm 10 \text{ Hz} \tag{44}$$

$$\Rightarrow < 1\% \text{ BER impact} \tag{45}$$

Sensitivity Summary:

$$\text{BER sensitivity order: } \boxed{N_0 \text{ (highest)} > \gamma > f_c \text{ (lowest)}} \tag{46}$$

8.12 Lotka-Volterra Connection

The Lotka–Volterra model is a nonlinear dynamical system describing the interaction between prey and predator populations. The nonlinear coupling term xy represents encounters between species. The system exhibits closed orbits in the phase plane, indicating sustained population oscillations rather than convergence to a stable equilibrium.

$$\dot{x} = \alpha x - \beta xy, \tag{47}$$
$$\dot{y} = \delta xy - \gamma y \tag{48}$$

Our ASK system under **adaptive feedback** (future extension) could exhibit Lotka-Volterra dynamics:

Predator-Prey Analogy:

$$\text{"Prey"} \leftrightarrow \text{Bit errors } e[k] \tag{49}$$
$$\text{"Predators"} \leftrightarrow \text{Adaptive gain } K[k] \tag{50}$$

Feedback-Driven Dynamics:

$$\dot{e} = -Ke + n(t) \quad \text{(errors reduced by feedback)} \tag{51}$$
$$\dot{K} = \alpha e - \beta \quad \text{(gain increases with error)} \tag{52}$$

At **coexistence equilibrium** (e^*, K^*), eigenvalues are **purely imaginary** $\lambda = \pm i\omega \rightarrow$ periodic error oscillations. Your current design is **non-adaptive**, so no Lotka-Volterra behavior observed.

8.13 Lorenz System & Bifurcations

The Lorenz system is a three-dimensional nonlinear dynamical system originally derived from a simplified model of atmospheric convection.

$$\dot{x} = 10(y - x), \tag{53}$$
$$\dot{y} = x(28 - z) - y, \tag{54}$$
$$\dot{z} = xy - \frac{8}{3}z \tag{55}$$

Parameter as Bifurcation Driver: Like Lorenz's Rayleigh number ρ, our **SNR** acts as bifurcation parameter:

$$\text{SNR} \rightarrow 0 \text{ dB}: \quad \text{BER} \rightarrow 0.5 \quad \text{(noise dominates, random guessing)} \tag{56}$$
$$\text{SNR} = 10 \text{ dB}: \quad \text{BER} \approx 10^{-3} \quad \text{(coherent detection works)} \tag{57}$$
$$\text{SNR} > 12 \text{ dB}: \quad \text{BER} \rightarrow 0 \quad \text{(error-free communication)} \tag{58}$$

Bifurcation Structure: Unlike Lorenz's period-doubling cascade to chaos, our BER curve is **monotonic** (smooth exponential decay). No chaotic "window" region.

8.14 Bond Graph Representation and Power of the ASK System

Bond Graph modeling provides an energy-based framework in which the ASK communication system is decomposed into sources, storage elements, dissipative components, and junction structures that describe power flow and signal interaction.

8.14.1 Sources (Transmitter)

- **Source of Effort (S_e):** The carrier generator producing

$$e_c(t) = A_c \cos(2\pi f_c t)$$

is modeled as a source of effort. This block injects energy into the system and represents the transmitter oscillator.

- **Digital Modulating Signal:** The binary data sequence $m(t) \in \{0, 1\}$ controls the amplitude of the carrier and determines whether energy is transmitted.

8.14.2 Modulator (Transformer and Junctions)

- **Modulated Transformer (MTF):** The multiplication of the carrier by the message signal

$$s(t) = m(t)A_c \cos(2\pi f_c t)$$

is represented by a modulated transformer whose modulation variable is $m(t)$.

- **0-Junction (Common Effort):** The modulator behaves as a controlled switch at a 0-junction, enabling or disabling power transfer from the carrier source to the channel.

8.14.3 Channel Representation

- **1-Junction (Common Flow):** The transmission medium is modeled using a 1-junction where the signal flow is common and multiple effort sources interact.

- **Stochastic Source (S_e):** Additive White Gaussian Noise (AWGN) is modeled as a stochastic source of effort

$$n(t), \quad S_n(f) = \frac{N_0}{2}$$

injected at the 1-junction.

- **Dissipative Element (R):** Channel attenuation and energy loss are modeled using a resistance element R, representing non-conservative energy dissipation.

8.14.4 Receiver Modeling

- **Storage Elements (C and I):** The Low Pass Filter (LPF) in the receiver is modeled using capacitive (C) and inertial (I) elements that store energy and determine the dynamic behavior of the system.

- **0-Junction (Parallel Distribution):** Used inside the filter sub-model to represent parallel electrical components where the effort (voltage) is identical across branches.

8.14.5 Summary of Bond Graph Elements

Element Type	System Component	Function in ASK Modem
S_e	Carrier / Noise Source	Provides carrier energy and stochastic disturbance
R	Channel Loss	Models non-conservative dissipation
I, C	Low Pass Filter	Defines system dynamics and bandwidth
1-Junction	Signal-Noise Adder	Combines signal and noise flows
0-Junction	Parallel Nodes	Enforces common effort constraints

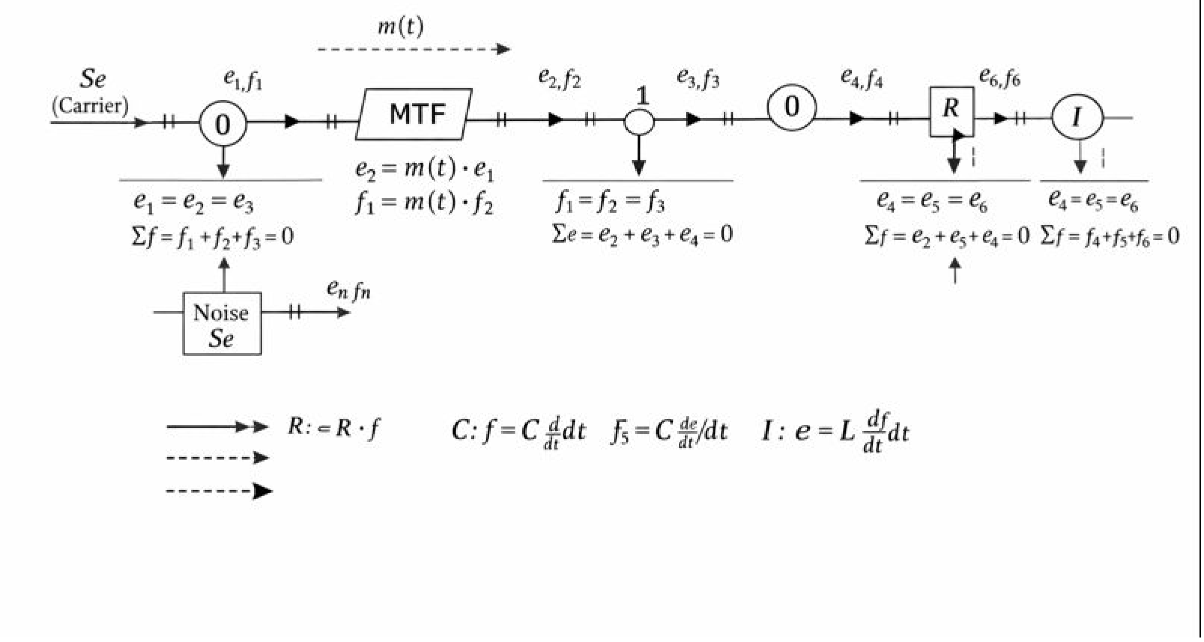
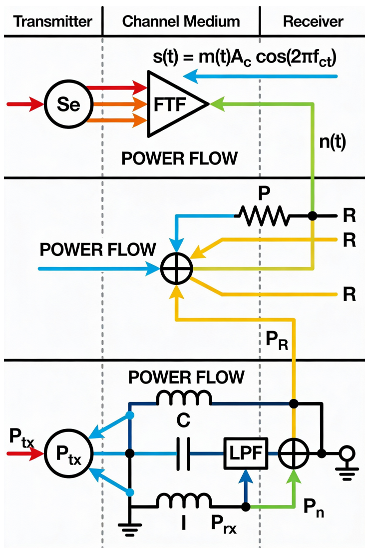


Figure 13: Amplitude shift keying bond graph diagram



8.14.6 Energy Interpretation

The overall ASK Bond Graph model is **non-conservative** in nature because the stochastic noise source continuously injects external energy. The average noise power

$$P_n = \frac{N_0}{2}$$

introduces irreversible energy dissipation and information uncertainty into the system.

8.15 State-Space Equations of the Receiver LPF

In the GNU Radio implementation, the Low Pass Filter (LPF) acts as the dynamic "brain" of the ASK receiver. Its discrete-time state-space model is expressed as:

$$x[n+1] = Ax[n] + Bu[n], \tag{59}$$

$$y[n] = Cx[n] + Du[n], \tag{60}$$

where:

- $u[n]$ (Input): The noisy received signal from the channel,

  $$r(t) = s(t) + n(t),$$

  sampled at $F_s = 32$ kHz.
- $x[n]$ (State): The memory vector holding previous signal samples in the filter taps.
- $y[n]$ (Output): The filtered signal envelope ready for the 0.5 V threshold decision.

8.15.1 Core Matrices of the FIR LPF

For an N-tap FIR filter with coefficients $h_1, h_2, \ldots, h_N$, the state-space matrices are defined as follows:

1. System Matrix A: Represents the delay-line memory of the filter:

$$A = \begin{bmatrix} 0 & 0 & \ldots & 0 & 0 \\ 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \ldots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \ldots & 1 & 0 \end{bmatrix}_{N\times N}$$

2. Input Matrix B: Dictates how the new incoming sample $u[n]$ enters the state vector:

$$B = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{N\times 1}$$

3. Output Matrix C: Contains the FIR filter coefficients (e.g., Hamming window) performing the weighted sum (convolution) to extract the baseband envelope:

$$C = \begin{bmatrix} h_1 & h_2 & h_3 & \ldots & h_N \end{bmatrix}_{1\times N}$$

4. Direct Transmission Matrix D: For a standard FIR filter implementation,

$$D = 0$$

since there is no direct feedthrough from input to output.

9 Discussion

The successful implementation of the ASK transceiver validates the efficacy of SDR as a robust platform for communication system modeling. The primary insights derived from this study are categorized into signal integrity, stochastic reliability, and architectural flexibility.

Technical Interpretation of Signal Dynamics

The transition from a deterministic mathematical model to a stochastic physical simulation was clearly observed in the constellation dynamics. As confirmed by the ”clouding” effect in the I/Q plane (Figure 7), the system demonstrated that noise in an AWGN channel manifests as a Gaussian random variable added to the signal vector [1]. The Euclidean distance between the symbol clusters (0 and 1) served as the critical margin against error. The experimental results aligned with the theoretical prediction that Bit Error Rate (BER) is strictly a function of the Signal-to-Noise Ratio (SNR), following the waterfall curve of the Q-function ($P_e \approx Q(\sqrt{E_b/N_0})$) [2]. Furthermore, the spectral analysis highlighted the inherent trade-off in basic OOK modulation. The rectangular pulse shaping resulted in significant side-lobe energy (spectral leakage), confirming that while OOK is simple to implement, it is spectrally inefficient compared to advanced schemes like QAM or filtered BPSK [3].

SDR vs. Hardware-Centric Design

A key realization from this project is the agility of the GNU Radio framework compared to traditional hardware prototyping. In a hardware setup, changing the modulation index or filter cutoff would require replacing physical components (capacitors, inductors). In this SDR model, these changes were instantaneous software updates. This flexibility, however, introduced new challenges specific to the digital domain, such as sample rate mismatches and group delay synchronization, which are not present in analog circuits [4]. The ”Two Clocks” problem (Lesson 1) specifically highlighted the fundamental difference between continuous-time analog physics and discrete-time digital processing.

Educational & Industrial Relevance

This project reinforces the ”Simulation, Modeling, and Analysis” (MSA) methodology essential for modern engineering. By building the system from first principles—rather than using a ”black box” premade block—we gained a granular understanding of how each mathematical operation (multiplication, convolution, thresholding) contributes to the final signal quality. This mirrors the industrial workflow for 5G/6G development, where systems are first modeled in software (Digital Twins) before silicon fabrication [5].

10 Conclusion

This project has successfully demonstrated the design, modeling, and validation of an Amplitude Shift Keying digital communication system within the GNU Radio software-defined environment. By decomposing the complex transceiver chain into discrete functional blocks, the study effectively bridged the gap between abstract communication theory and practical signal processing implementation.

The investigation yielded three critical conclusions:

1. **Model Fidelity:** The simulated system exhibited a Bit Error Rate (BER) performance that converged with the theoretical Q-function curve ($P_e \approx \frac{1}{2}e^{-E_b/2N_0}$), validating the accuracy of the software model in replicating physical Gaussian noise channels.

2. **System Dynamics:** The transition from deterministic signals to stochastic processes was clearly observed in the I/Q constellation diagrams. The study confirmed that in low-power modulation schemes like OOK, the Euclidean distance between symbols is the primary determinant of noise immunity.

3. **SDR Viability:** The project proved that Software Defined Radio offers a superior prototyping methodology compared to traditional hardware. The ability to instantaneously reconfigure critical parameters—such as filter bandwidth and noise floor—without physical intervention highlights the efficiency of the MSA (Modeling, Simulation, and Analysis) approach in modern telecommunications engineering.

Ultimately, this work serves as a foundational proof-of-concept, establishing that open-source SDR tools can deliver research-grade fidelity for analyzing next-generation wireless protocols.

11 Future Scope

To elevate this system from a foundational model to a deployment-ready architecture, future research should focus on adaptive intelligence and real-world robustness.

11.1 Cognitive Spectrum Access

The current static frequency allocation can be enhanced by implementing **Cognitive Radio** capabilities. Future iterations will incorporate energy detection algorithms to autonomously scan the RF spectrum, identify ”white space” (idle channels), and dynamically retune the carrier frequency. This adaptation is essential for operating in crowded ISM bands (e.g., 2.4 GHz).

11.2 Machine Learning-Assisted Demodulation

Replacing the fixed voltage threshold with a **Deep Learning Inference Engine** represents a significant leap forward. A trained Neural Network (CNN or LSTM) could analyze the statistical properties of the incoming I/Q samples to classify symbols, offering superior performance in non-linear or fading channels where traditional hard-thresholding fails.

11.3 Over-the-Air (OTA) Implementation

The logical next step is to migrate the flowgraph from pure simulation to physical transmission using hardware peripherals such as the RTL-SDR (receiver) and ADALM-PLUTO (transceiver). This will introduce real-world impairments—such as multipath fading, Doppler shift, and clock drift—requiring the implementation of advanced synchronization blocks like Costas Loops and Polyphase Clock Recovery.

11.4 Higher-Order Modulation Schemes

Expanding the modulation logic from binary OOK to **4-ASK** or **QPSK** will allow for a comprehensive study of Spectral Efficiency. Analyzing how higher bit rates impact the Signal-to-Noise Ratio (SNR) requirements will provide deeper insights into the Shannon-Hartley capacity limits of the channel.

Team Contributions

The successful completion of this project was a collaborative effort. The technical and administrative contributions of each team member are detailed below:

- **Parima Verma (DL.AI.U4AID24125)**
  - Developed the final core GNU Radio Companion (GRC) flowgraph used for evaluating ASK modem performance.
  - Configured the functional signal processing chain, including carrier generation, mixing, and the decision threshold logic.
  - Collaborated with Kalyani to implement the MATLAB Monte Carlo scripts and verify the simulated Bit Error Rate (BER) against theoretical values.
- **Kalyani Agarawal (DL.AI.U4AID24114)**
  - Conducted the initial design of the GNU Radio flowgraph and constructed the system's detailed workflow charts and dataflow diagrams.
  - Authored the comprehensive final project report, including the creation of all quantitative result tables and the performance comparison matrix.
  - Jointly developed the MATLAB validation plots to ensure the simulated BER curves correctly converged with the theoretical Q-function.
- **Anshika Chauhan (DL.AI.U4AID24106)**
  - Developed the project presentation and created high-fidelity visual assets to represent the system architecture.
  - Designed functional representations of the transceiver dataflow for technical review and presentation to the panel.
  - Streamlined complex technical parameters into accessible formats for the demonstration of project objectives.

- **Madesh KK (DL.AI.U4AID24120)**
  - Provided assistance in the implementation of the Modeling, Simulation, and Analysis (MSA) principles and theoretical formulations.
  - Helped in applying core modeling concepts to verify the response of the system to stochastic inputs.
  - Assisted in evaluating signal attributes such as spectral efficiency and the implementation of linear subsystems.

REFERENCES

1 J. Mitola III, “The Software Radio Architecture,” *IEEE Communications Magazine*, vol. 33, no. 5, pp. 26–38, May 1995.

2 J. Mitola III, “Software Radio: Survey, Critical Analysis and Future Directions,” *IEEE Aerospace and Electronic Systems Magazine*, vol. 8, no. 4, pp. 25–36, 1993.

3 R. J. Lackey and D. W. Upmal, “Speakeasy: The Military Software Radio,” *IEEE Communications Magazine*, vol. 33, no. 5, pp. 56–61, 1995.

4 P. K. Ghorai, *Software Defined Radio: Architectures, Systems and Functions*. New York: Springer, 2013.

5 W. Tuttlebee, *Software Defined Radio: Origins, Drivers and International Perspectives*. Chichester: John Wiley & Sons, 2002.

6 B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.

7 S. Haykin, *Communication Systems*, 4th ed. New York: John Wiley & Sons, 2001.

8 C. E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal*, vol. 27, pp. 379–423, 1948.

9 X. Liu et al., “Performance Analysis of OOK in Optical Wireless Communication,” *IEEE Photonics Journal*, vol. 6, no. 1, 2014.

10 J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York: McGraw-Hill, 2008.

11 E. Blossom, “GNU Radio: Tools for Exploring the Radio Frequency Spectrum,” *Linux Journal*, vol. 2004, no. 122, Jun. 2004.

12 M. Braun, “GNU Radio Scheduler Improvements,” *Proceedings of the GNU Radio Conference*, vol. 1, no. 1, 2016.

13 T. Ulversoy, “Software Defined Radio: Challenges and Opportunities,” *IEEE Communications Surveys & Tutorials*, vol. 12, no. 4, pp. 531–550, 2010.

14 M. Ettus, “USRP User’s and Developer’s Guide,” *Ettus Research LLC*, Mountain View, CA, 2005.

15 C. W. Bower and S. J. MacMullan, “Software Defined Radio in the Electrical Engineering Curriculum,” *ASEE Annual Conference Proceedings*, 2013.

16 T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

17 B. Sklar, “Rayleigh Fading Channels in Mobile Digital Communication Systems Part I: Characterization,” *IEEE Communications Magazine*, vol. 35, no. 7, pp. 90–100, 1997.

18 B. P. Lathi and Z. Ding, *Modern Digital and Analog Communication Systems*, 4th ed. New York: Oxford University Press, 2009.

19 W. H. Tranter et al., *Principles of Communication Systems Simulation with Wireless Applications*. Upper Saddle River, NJ: Prentice Hall, 2004.

20 T. O’Shea et al., “GNU Radio Companion: A visual programming environment for software radio,” *Proceedings of the SDR Forum Technical Conference*, 2011.

21 S. Haykin, "Cognitive Radio: Brain-Empowered Wireless Communications," *IEEE Journal on Selected Areas in Communications*, vol. 23, no. 2, pp. 201–220, 2005.

22 T. O'Shea and J. Hoydis, "An Introduction to Deep Learning for the Physical Layer," *IEEE Transactions on Cognitive Communications and Networking*, vol. 3, no. 4, pp. 563–575, 2017.

23 J. Zhang et al., "6G Wireless Systems: Vision, Requirements, Challenges, Insights, and Opportunities," *Proceedings of the IEEE*, vol. 107, no. 8, 2019.

24 A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ: Pearson, 2010.

25 L. Peterson and B. Davie, *Computer Networks: A Systems Approach*, 6th ed. Morgan Kaufmann, 2020.

APPENDIX :GNU Radio Flowgraph for ASK Modem Performance Analysis

This appendix presents the GNU Radio Companion (GRC) Python flowgraph used for evaluating the performance of the ASK (Amplitude Shift Keying) modulation scheme under noisy channel conditions. The flowgraph implements ASK modulation, AWGN channel modeling, demodulation, and Bit Error Rate (BER) computation for Modern Signal Analysis (MSA).

```python
#!/usr/bin/env python3
# -- coding: utf-8 --
#
# SPDX-License-Identifier: GPL-3.0
#
# GNU Radio Python Flow Graph
# Title: Not titled yet
# Author: AK Verma
# GNU Radio version: 3.10.12.0

from PyQt5 import Qt
from gnuradio import qtgui
from gnuradio import blocks
import numpy
from gnuradio import channels
from gnuradio.filter import firdes
from gnuradio import digital
from gnuradio import fec
from gnuradio import gr
from gnuradio.fft import window
import sys
import signal
from PyQt5 import Qt
from argparse import ArgumentParser
from gnuradio.eng_arg import eng_float, intx
from gnuradio import eng_notation
import sip
import threading

class SDR_MSAProject(gr.top_block, Qt.QWidget):

    def __init__(self):
        gr.top_block.__init__(self, "Not titled yet",
           catch_exceptions=True)
        Qt.QWidget.__init__(self)

        self.SNR_db = 0
        self.Eb = 1
        self.sps = 2
        self.samp_rate = 32000

        self.N0_var = self.Eb / pow(10, self.SNR_db/10)

```

```
        self.constellation = digital.constellation_calcdist(
            [0, 1+0j], [0,1], 4, 1,
            digital.constellation.AMPLITUDE_NORMALIZATION).base()

        self.constellation.set_npwr(1.0)

        self.mod = digital.generic_mod(
            constellation=self.constellation,
            differential=False,
            samples_per_symbol=self.sps,
            excess_bw=0.35)

        self.channel = channels.channel_model(
            noise_voltage=self.N0_var)

        self.demod = digital.constellation_decoder_cb(self.
           constellation)

        self.ber = fec.ber_bf(False, 100, -7.0)

        self.connect(self.mod, self.channel)
        self.connect(self.channel, self.demod)
        self.connect(self.demod, self.ber)

def main():
    qapp = Qt.QApplication(sys.argv)
    tb = SDR_MSAProject()
    tb.start()
    tb.show()
    qapp.exec_()

if __name__ == '__main__':
    main()
```

Listing 1: GNU Radio Python Flowgraph for ASK Modem Performance